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FACULTY OF
ENGINEERING SCIENCE

Topology Optimization of Heat Transfer Applications Operating in the Turbulent Regime

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Master of Science in Mechanical
Engineering

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Preface

This thesis marks the culmination of my Master of Science in Mechanical Engineering at KU Leuven. The journey of exploring computational fluid dynamics and mathematical optimization has been both intensely challenging and deeply rewarding. Bridging the gap between theoretical mathematics and practical engineering applications required countless hours of coding, debugging, and analyzing flow fields, but seeing the optimizer successfully carve out organic, aerodynamic fluid channels made the entire process worthwhile.

I would like to express my sincere gratitude to my supervisors, Prof. Niels Horsten and Prof. Maarten Blommaert, for providing me with the opportunity to work on this fascinating topic. Their academic guidance, critical insights, and continuous support were instrumental in shaping the direction and rigor of this research. A very special thanks goes to my assistant-supervisor, Esteban Foronda Obando, for his day-to-day patience, technical expertise, and willingness to help me navigate the complex numerical hurdles within FEniCS.

Furthermore, I would like to thank the assessors, Prof. Elke Deckers and Olivier Renders, for their time and valuable feedback in evaluating this work.

Finally, I want to thank my family and friends for their unwavering encouragement, patience, and perspective throughout the course of my studies.

Tiebert Lefebure

Contents

Abstract

As modern electronic and automotive systems grow increasingly compact and powerful, efficient thermal management has emerged as a fundamental engineering challenge. While topology optimization has proven to be a highly effective computational design tool for the fluid manifolds that drive these cooling systems, traditional methodologies rely heavily on laminar flow approximations. These laminar assumptions severely limit industrial applicability, as they fail to capture the separation, curvature effects, and turbulence-model dependence inherent to high-speed internal flows.

This thesis develops and implements a density-based topology optimization framework for turbulent internal flows as a baseline for future heat-transfer applications. The computational methodology is built upon the open-source FEniCS platform and utilizes the fictitious domain method with Brinkman penalization to conceptualize solid boundaries within a continuous fluid mesh. To capture the complex physics of high-Reynolds flows, the framework integrates the one-equation Spalart-Allmaras (SA) turbulence model. The optimization process is primarily driven by a frozen-turbulence continuous adjoint, while an exploratory semi-frozen coupled adjoint is also examined.

The assessment strategy is deliberately separated into three stages. First, the predictive scope of the SA model for curvature-dominated internal flows is established and benchmarked against Ansys Fluent. Second, the custom SA implementation in FEniCS is verified on turbulent benchmark flows. Third, the final optimized geometries are exported from FEniCS and re-analyzed in Ansys Fluent. This final stage compares FEniCS-SA with Ansys-SA on the same geometry and then evaluates the same designs with SST k-ω and, secondarily, standard k-ϵ to quantify turbulence-model sensitivity. In this revised framing, laminar-versus-turbulent design comparison is retained as supporting context rather than treated as CFD validation in itself.

The resulting thesis is therefore intentionally conservative in its claims. Its central question is not merely whether turbulent topologies look different from laminar ones, but whether SA-driven topology optimization produces low-loss designs whose predicted performance remains credible under independent CFD re-analysis. This establishes a more defensible and transparent baseline for future three-dimensional and conjugate heat-transfer extensions.

List of Figures and Tables

List of Figures

List of Tables

List of Abbreviations and Symbols

Abbreviations

CAD	Computer-Aided Design
CEV	Constant Eddy Viscosity
CFD	Computational Fluid Dynamics
CG1	Continuous Galerkin (Linear Basis Functions)
CHT	Conjugate Heat Transfer
DG0	Discontinuous Galerkin (Piecewise Constant Elements)
DNS	Direct Numerical Simulation
FEM	Finite Element Method
FVM	Finite Volume Method
IPCS	Incremental Pressure Correction Scheme
MMA	Method of Moving Asymptotes
PDE	Partial Differential Equation
RAMP	Rational Approximation of Material Properties
RANS	Reynolds-Averaged Navier-Stokes
SA	Spalart-Allmaras
SIMP	Solid Isotropic Material with Penalization
SST	Shear Stress Transport
SUPG	Streamline Upwind Petrov-Galerkin
TO	Topology Optimization
UFL	Unified Form Language

Symbols

Fluid Dynamics & Turbulence

d, y	Distance to the nearest wall / Hydraulic diameter [m]
De	Dean number [-]
k	Turbulent kinetic energy [m^2/s^2]
$p, \bar{p}, p'$	Instantaneous, mean, and fluctuating static pressure [Pa]
ΔP	Macroscopic total pressure drop [Pa]
R_c	Radius of curvature [m]
Re	Bulk Reynolds number [-]
$S, \tilde{S}$	Scalar vorticity magnitude and modified SA vorticity magnitude [1/s]
t	Time / Pseudo-time [s]
$\mathbf{u}, \bar{u}_i$	Velocity vector / Mean velocity component [m/s]
u'_i	Fluctuating velocity component [m/s]
u_τ	Friction velocity [m/s]
y^+	Non-dimensional wall distance [-]
μ, μ_t	Dynamic and turbulent eddy viscosity [Pa·s]
ν, ν_t	Kinematic and turbulent eddy kinematic viscosity [m^2/s]
$\tilde{\nu}$	Spalart-Allmaras working variable [m^2/s]
ρ	Fluid density [kg/m^3]
Ω_{ij}	Vorticity tensor [1/s]
κ	von Kármán constant [-]

Numerical Implementation (FEniCS)

G, G_0	Relaxed Eikonal field variable and boundary value [1/m]
h, h_{avg}	Finite element cell diameter and average cell size [m]
$\|\| \cdot \|\|_{L_2}$	L_2 (Euclidean) norm [-]
$[\![\cdot]\!]$	Jump operator across an internal element facet [-]
$\mathbf{n}$	Outward-pointing normal vector [-]
v, ξ	Test functions for pressure and turbulence equations [-]
$\alpha_u, \alpha_{\tilde{\nu}}$	Relaxation factors for velocity and turbulence [-]
Δt	Pseudo-time step [s]
ϵ_{tol}	Convergence tolerance [-]
ϕ	General scalar field variable [-]
σ_w	Eikonal relaxation constant [-]
τ	SUPG stabilization time parameter [s]

Topology Optimization

$\alpha(\gamma)$	Continuous Brinkman penalization term / inverse permeability
$\alpha_{solid}, \alpha_{fluid}$	Maximum solid and minimum fluid Brinkman penalties
α_{dg}	Artificial diffusion scaling factor for DG0 filter [-]
β	Heaviside projection steepness parameter [-]
γ, γ_{raw}	Continuous and raw design variable (1 for fluid, 0 for solid) [-]
$\tilde{\gamma}, \bar{\gamma}$	Filtered and Heaviside-projected design variables [-]
η	Heaviside projection threshold [-]
J	Objective function (Power Dissipation) [W]
q	RAMP penalization tuning parameter [-]
r	Continuous Helmholtz filter radius [m]
$\mathbf{s}, \mathbf{s}^*$	Monolithic forward state and continuous adjoint state vectors
V_f, V_{domain}	Physical fluid volume and total design domain volume [m^3]
V_{max}	Maximum allowable fluid volume fraction constraint [-]
$\Omega, \partial\Omega, \Gamma_{int}$	Computational domain, domain boundary, and internal element facets

Chapter 1

Introduction

1.1 Background and Motivation

With the continued miniaturization of components in the electronics and automotive industries, thermal management has become a critical bottleneck in the development of new technologies. While the heat sinks themselves are vital, the efficiency of any cooling system relies heavily on the fluid manifolds that deliver the coolant. If the flow distribution is uneven or if pumping the fluid requires excessive power due to massive pressure drops, the overall efficiency of the system severely degrades. Addressing this fluid delivery problem from an engineering perspective is a challenging task due to the complex aerodynamic behaviors that govern high-speed flow networks.

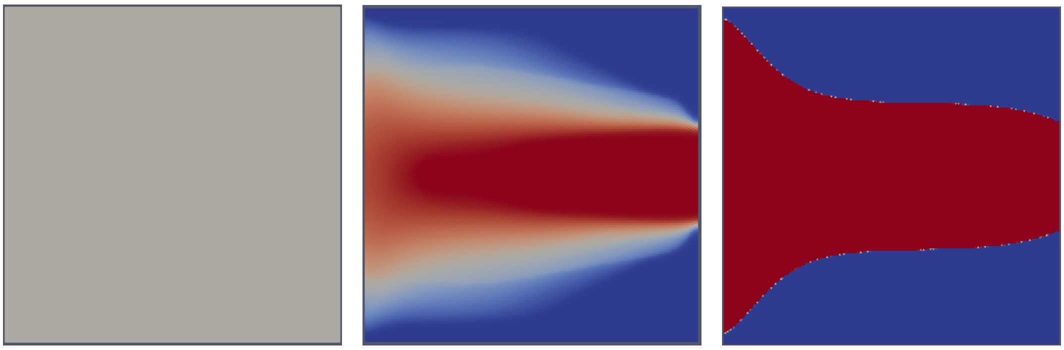
Innovative design strategies based on mathematical optimization have made it possible to improve the performance of these devices while capturing the complexity of the underlying physical phenomena. Among these strategies, topology optimization has emerged as a highly effective computational design tool for advanced thermal management [20]. For instance, the geometry of a complex microfluidic channel network can be automatically optimized to distribute coolant with specified flow rates without relying on human intuition [52]. Furthermore, the ultimate frontier of this technology lies in the design of conjugate and two-fluid heat exchangers, where the algorithm must simultaneously maximize heat transfer while strictly limiting the fluid pumping power [27]. However, before multi-physics heat transfer can be reliably optimized at an industrial scale, the fundamental algorithms governing the turbulent fluid dynamics must first be stabilized and validated.

1.2 Conceptual Overview of Density-Based Topology Optimization

To understand the challenges of applying optimization to turbulent fluids, it is first necessary to distinguish topology optimization from traditional design methodologies. Historically, computational design relied on size optimization (altering the thickness of predefined components) or shape optimization (moving the boundary nodes of a predefined geometry). While effective, these methods are inherently limited by their starting topology. A shape optimization algorithm cannot spontaneously create a new hole or split a single fluid channel into two.

Density-based topology optimization, pioneered by Bendsøe and Kikuchi in 1988 for solid mechanics [14], approaches the problem fundamentally differently. Instead of moving boundaries, the design space (the computational domain) is discretized into a fixed grid of finite elements. The algorithm then determines the optimal distribution of material across this grid. In fluid problems, this is achieved by assigning a continuous design variable, $\gamma \in [0, 1]$, to every element in the domain. In the specific convention adopted for this thesis, a value of $\gamma = 1$ represents a pure fluid channel (e.g., the cooling air or water), while $\gamma = 0$ represents a solid, impermeable structure (e.g., a manifold wall).

However, because gradient-based optimizers require continuous and differentiable functions, this discrete binary choice is mathematically relaxed. The design variable is allowed to take intermediate "gray" values between 0 and 1. To give physical meaning to these intermediate states, the fluid equations are modified using a fictitious porous medium approach, where intermediate densities correspond to varying levels of flow permeability. The optimization algorithm iteratively updates the γ field (as conceptualized in Figure 1.1) to minimize a specific objective function (e.g., dissipated power) subject to physical constraints (e.g., a maximum allowable fluid volume fraction). The choice of the initial value for γ is a critical parameter for solver stability. A uniform initial guess of $\gamma = 0.5$ is standard practice to ensure an unbiased algorithmic search across the design space.



(A) Initial State ($\gamma = 0.5$) (B) Continuous Distribution (C) Discrete Topology

FIGURE 1.1: Iterative density-based topology optimization process: (a) initialization with a uniform field, (b) intermediate continuous distribution where $\gamma \in [0, 1]$, and (c) final discrete fluid topology where $\gamma \in \{0, 1\}$.

1.3 The Evolution of Fluid Topology Optimization

While density-based topology optimization became a standard tool in structural engineering in the late 1990s, its application to fluid dynamics is relatively recent.

The optimization of fluid manifolds remains an active and highly challenging area of research [20]. The genesis of fluid topology optimization is widely attributed to Borrvall and Petersson (2003), who successfully applied the technique to minimize power dissipation in creeping Stokes flow [15]. This was quickly followed by Gersborg-Hansen et al. (2005), who extended the framework to steady, laminar Navier-Stokes flows [22]. Over the past decade, the literature surrounding laminar flow optimization expanded extensively, successfully adapting the methodology to standard incompressible Navier-Stokes equations [17, 4, 52].

However, the assumption of laminar flow significantly simplifies the governing equations and the subsequent derivation of the adjoint sensitivities required for gradient-based optimization. Unfortunately, this assumption prevents the application of these methodologies to most industrial and high-performance applications, where flow velocities are high and turbulence dominates the flow physics. Only recently has the computational mechanics community begun to tackle the severe non-linearities and numerical instabilities associated with optimizing geometries directly in the fully turbulent regime [49, 18, 50].

Extending topology optimization to the turbulent regime is highly non-trivial. It requires coupling optimization algorithms with complex Reynolds-Averaged Navier-Stokes (RANS) turbulence models. These models introduce steep velocity gradients, highly non-linear behaviors, and severe numerical instabilities during the iterative optimization process. To mitigate this, gradient-based turbulent optimizations frequently rely on the "frozen turbulence" assumption, where the turbulent viscosity field is held constant during the sensitivity analysis to avoid differentiating the complex turbulence transport equations. While this assumption provides necessary solver stability and has driven massive success in industrial shape optimization for automotive aerodynamics [34], advancing toward fully coupled turbulent sensitivities in topology optimization remains an ongoing and highly complex mathematical challenge.

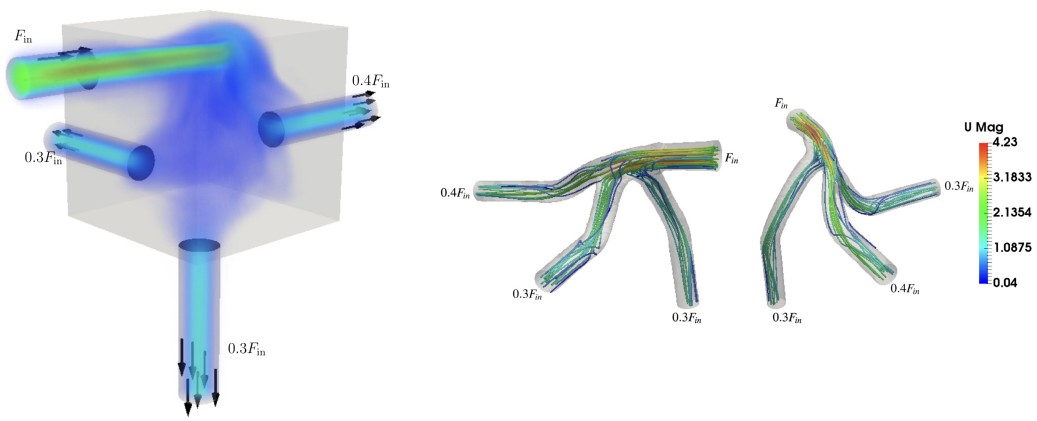


FIGURE 1.2: Topology optimized manifolds with turbulent flow. Adapted from Dilgen et al. [18].

Pioneering works by Yoon (2016) [49] and Dilgen et al. (2018) [18] have pushed the boundaries of these turbulent frameworks. As seen in Figure 1.2, integrating advanced turbulence models allows the optimizer to generate highly complex, organic fluid manifolds specifically tailored to manipulate and guide turbulent flow distributions efficiently.

1.4 Objectives and Research Questions

The main objective of this Master's thesis is to extend the application of topology optimization to the design of fluid manifolds operating strictly in the turbulent regime. To achieve this goal, this work investigates the coupling of fluid dynamics and turbulence modeling within an automated mathematical optimization framework. By strictly isolating the fluid mechanics from complex thermodynamic calculations, this thesis bounds its scope to establish a highly stable and mathematically rigorous baseline for pure aerodynamic manifold design.

Rather than relying solely on black-box commercial Computational Fluid Dynamics (CFD) software, the core of this thesis involves implementing the framework using **FEniCS**, an open-source computing platform for solving partial differential equations [5]. FEniCS, combined with high-level discrete adjoint methods (e.g., `dolfin-adjoint` [6]), provides the mathematical flexibility required to implement, verify, and integrate the one-equation Spalart-Allmaras (SA) turbulence model into a fluid-design loop. For full transparency and reproducibility, the complete FEniCS source code developed for this thesis is open-source and publicly available at: `https://github.com/TiebertLefebure/TurbulentTO.git`.

At this stage, it is important to distinguish two different comparisons that appear later in the thesis. A comparison between laminar-optimized and turbulent-optimized layouts is useful as supporting design context, because it shows how inertia and turbulence alter the preferred topology. However, this comparison is not treated as the principal CFD validation step. The decisive assessment instead consists of re-analyzing the final SA-optimized geometries in Ansys Fluent, first with the same SA model and then with alternative RANS closures.

Based on this methodology, this thesis seeks to answer the following specific research questions:

1. **How accurately does the Spalart-Allmaras turbulence model predict curvature-dominated internal turbulent flows, and where do its physical limitations appear?**

2. **Can a custom standard-Galerkin FEniCS implementation of the Spalart-Allmaras turbulence model converge robustly and achieve quantitative agreement with established commercial CFD solvers?**

3. **To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys re-analysis, both with the same SA model and with alternative RANS closures?**

1.5 Outline of the Thesis

The remainder of this thesis is structured into the following five chapters:

- **Chapter 2 (Turbulence Modeling):** Provides the theoretical background on fluid dynamics and turbulence. It discusses the Navier-Stokes equations, the principles of RANS modeling, and the physical characteristics of the Spalart-Allmaras model.

- **Chapter 3 (Implementation and Verification of the Spalart-Allmaras Model in FEniCS):** Details the numerical implementation of the chosen turbulence model within the FEniCS framework, including fractional-step solvers, standard Galerkin discretization, convergence controls, and quantitative benchmark validations.

- **Chapter 4 (Topology Optimization for Turbulent Flows: Methodology):** Introduces the density-based optimization mathematical framework. It details the Brinkman penalization, the continuous adjoint derivations, the rationale for adopting the frozen turbulence assumption as the baseline design engine, and the exploratory coupled formulation.

- **Chapter 5 (Topology Optimization for Turbulent Flows: Results, Cross-Code Verification, and Model Sensitivity):** Presents the turbulent topology optimization results, documents the export of the final designs to Ansys Fluent, compares FEniCS-SA against Ansys-SA on the same geometries, and studies sensitivity to alternative turbulence models such as SST k-ω and standard k-ϵ.

- **Chapter 6 (Conclusion and Future Work):** Summarizes the main findings, answers the research questions conservatively, and outlines the remaining limitations, with particular emphasis on turbulence-model dependence, geometry extraction, and future multi-physics heat-transfer applications.

Chapter 2

Turbulence Modeling

2.1 Introduction to Fluid Dynamics and Turbulence

Fluid dynamics is governed by the Navier-Stokes equations, which describe the conservation of mass and momentum for a viscous continuum fluid [47]. In this introductory text, these equations are presented using index notation (Einstein summation convention) to clearly illustrate the individual tensor components. In this convention, the indices i and j represent the three spatial dimensions (1, 2, and 3, corresponding to the x, y, and z axes). Consequently, x_i denotes the Cartesian coordinates, and u_i represents the corresponding components of the velocity vector. A repeated index within a single term implies a mathematical summation over all three spatial directions. (Note: In subsequent chapters dealing with finite element implementation, this notation will shift to standard vector calculus to align with FEniCS unified form language syntax). The fundamental principle of mass conservation is expressed by the continuity equation. For a general, compressible fluid, this is defined as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u_i)}{\partial x_i} = 0 \tag{2.1}$$

where ρ is the fluid density. However, for most liquid cooling applications and low-speed fluid manifolds, the fluid can be safely assumed to be incompressible. Under the incompressible flow assumption, the density ρ is constant in both space and time. Therefore, the time derivative of density becomes zero, and the constant density can be factored out of the spatial derivative. Dividing the equation by ρ yields the simplified, incompressible continuity equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{2.2}$$

Similarly, for an incompressible, Newtonian fluid with a constant dynamic viscosity μ, the conservation of momentum is expressed as:

$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} \right) \tag{2.3}$$

where p represents the static pressure of the fluid. In principle, solving Equation 2.3 directly yields the exact flow field. This approach is known as Direct Numerical Simulation (DNS). However, as the Reynolds number (Re) increases, the flow transitions into a chaotic turbulent state characterized by a broad spectrum of spatial and temporal scales [36]. Resolving the smallest turbulent eddies requires extremely fine spatial grids. For industrial applications, including the design of electronics cooling manifolds, DNS is computationally prohibitive. Therefore, engineers rely on turbulence modeling to predict the averaged effects of these fluctuations without resolving the complete turbulent spectrum [46, 7].

2.2 Reynolds-Averaged Navier-Stokes (RANS) Equations

The most common approach to turbulence modeling in engineering is the Reynolds-Averaged Navier-Stokes (RANS) framework, based on the theories originally introduced by Osborne Reynolds [36]. The core principle of RANS is the Reynolds decomposition. This mathematical operation separates the instantaneous flow variables into a time-averaged mean component ($\bar{u}_i$, $\bar{p}$) and a fluctuating component (u_i', p'):

$$u_i = \bar{u}_i + u_i' \quad \text{and} \quad p = \bar{p} + p' \tag{2.4}$$

Substituting these decomposed variables into the standard Navier-Stokes equations and taking the time average yields the RANS momentum equations:

$$\rho\left(\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j}\right) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu \frac{\partial \bar{u}_i}{\partial x_j} - \rho \overline{u_i' u_j'}\right) \tag{2.5}$$

The averaging process introduces a new term, $-\rho\overline{u_i' u_j'}$, known as the Reynolds stress tensor. This tensor represents the momentum transfer caused by turbulent fluctuations. The introduction of this term creates a fundamental mathematical closure problem because the RANS equations now contain more unknowns than available physical equations [46].

2.2.1 The Boussinesq Hypothesis and the Isotropic Assumption

To close the system and solve for the mean flow, most RANS models rely on the Boussinesq eddy viscosity hypothesis [36]. This hypothesis assumes that the turbulent momentum transfer is analogous to molecular momentum transfer. It relates the Reynolds stresses linearly to the mean velocity gradients using a proportionality constant called the turbulent eddy viscosity (μ_t):

$$-\rho\overline{u_i' u_j'} = \mu_t\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right) - \frac{2}{3}\rho k \delta_{ij} \tag{2.6}$$

where k is the turbulent kinetic energy and δ_{ij} is the Kronecker delta. The primary task of any Boussinesq-based turbulence model is to calculate this local eddy viscosity. It is crucial to highlight a fundamental physical limitation of Equation 2.6: the Boussinesq hypothesis inherently assumes that μ_t is an *isotropic* scalar quantity. It mathematically enforces that turbulence behaves identically in all spatial directions and aligns perfectly with the mean strain rate. While this assumption holds for simple shear flows, it systematically fails in predicting anisotropic phenomena, such as the strong secondary flows induced by streamline curvature.

2.3 Boundary Layer Theory and Near-Wall Treatment

Before selecting a specific turbulence equation to calculate μ_t, the physical structure of a turbulent boundary layer must be considered, as the model must accurately resolve these near-wall regions. The boundary layer is traditionally divided into three distinct regions based on the non-dimensional wall distance (y^+) and non-dimensional velocity (u^+):

$$y^+ = \frac{y u_\tau}{\nu}, \quad u^+ = \frac{\bar{u}}{u_\tau} \tag{2.7}$$

where y is the absolute distance from the wall, ν is the kinematic viscosity (μ/ρ), and $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity derived from the wall shear stress (τ_w) [36].

1. **The viscous sublayer ($y^+ < 5$):** Immediately adjacent to the wall, turbulent fluctuations are damped out by kinematic constraints. Viscous shear dominates, and the velocity profile is strictly linear ($u^+ = y^+$).

2. **The buffer layer ($5 < y^+ < 30$):** A transition region where both viscous and turbulent shear stresses are of equal magnitude.

3. **The log-law region ($y^+ > 30$):** Farther from the wall, turbulence dominates, and the velocity follows a logarithmic distribution predicted by the von Kármán constant [46].

High-Reynolds models (like the standard $k - \epsilon$ model) use empirical "wall functions" to bridge the gap between the wall and the log-law region, requiring the first computational mesh element to be placed at $y^+ > 30$. However, for topology optimization, where solid-fluid boundaries continuously evolve during the optimization loop, empirical wall functions introduce severe mathematical discontinuities and numerical instabilities. Therefore, a "wall-resolved" model that integrates the governing equations all the way through the viscous sublayer ($y^+ \approx 1$) is strictly required for robust gradient-based optimization.

2.4 The Spalart-Allmaras One-Equation Model

While two-equation models (such as $k-\epsilon$ or $k-\omega$) solve separate transport equations for the turbulent kinetic energy and a turbulent length scale, they introduce highly non-linear source terms. These terms frequently cause numerical instabilities during the adjoint sensitivity analysis required for topology optimization. To achieve a trade-off between physical accuracy and computational efficiency, this thesis utilizes the Spalart-Allmaras (SA) model, originally developed in 1992 for aerodynamic flows with adverse pressure gradients [39]. The SA model is a one-equation model that solves a single transport equation directly for a working variable, $\tilde{\nu}$, which is subsequently linked to the true kinematic eddy viscosity ($\nu_t = \mu_t/\rho$) [40]. The transport equation for the standard SA model is given by:

$$\frac{\partial \tilde{\nu}}{\partial t} + \bar{u}_j \frac{\partial \tilde{\nu}}{\partial x_j} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{d}\right)^2 + \frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left((\nu + \tilde{\nu})\frac{\partial \tilde{\nu}}{\partial x_j}\right) + c_{b2}\left(\frac{\partial \tilde{\nu}}{\partial x_j}\right)^2\right] \tag{2.8}$$

The terms on the right-hand side represent the production, destruction, and diffusion of the turbulent working variable, respectively. Crucially, the production term relies on a modified vorticity scalar, $\tilde{S}$. This indicates that turbulence generation in the SA model is fundamentally driven by fluid shear, a physical characteristic that introduces unique boundary-condition sensitivities during advanced adjoint calculations (discussed further in Chapter 4). Furthermore, the destruction term is explicitly a function of d, the absolute distance to the nearest wall. This makes the SA model a highly effective "wall-resolved" low-Reynolds model. It is inherently sensitive to near-wall damping effects and integrates smoothly through the viscous sublayer without requiring complex empirical wall functions or a second length-scale equation [37]. Detailed derivations of the model constants and wall damping functions are deferred to Chapter 3, where the numerical implementation in FEniCS is discussed.

2.5 Evaluation of Curvature Effects and the Dean Number

Despite its computational advantages for topology optimization, the standard SA model relies on a single length scale tied to the nearest wall distance. As it lacks an explicit rotation/curvature correction and remains constrained by the Boussinesq eddy-viscosity assumption, it struggles to capture the anisotropic physics of streamline curvature. When fluid passes through a bend, centrifugal forces induce secondary flows perpendicular to the main flow direction. To quantify the strength of these secondary flows, the dimensionless Dean number (De) is utilized:

$$De = Re_d \sqrt{\frac{d}{2R_c}} \tag{2.9}$$

where d is the characteristic length of the cross-section (e.g., the pipe diameter D or hydraulic diameter D_h), R_c is the radius of curvature, and Re_d is the bulk Reynolds number based strictly on that exact same characteristic length. A higher Dean number indicates a stronger cross-sectional influence of centrifugal forces.

A review of the literature demonstrates that the predictive capability of the SA model is heavily dependent on this Dean number. For mildly curved flows, the SA model performs relatively well. Apalowo [10] evaluated a 90° circular pipe bend with a Dean number of $De \approx 9,000$ (operating at $Re_D \approx 34,000$, based on the pipe diameter). Under these moderate conditions, where the bend radius was seven times larger than the pipe diameter ($R_c = 7D$), the SA model provided a highly acceptable approximation of the streamwise velocity profiles compared to higher-order models, successfully capturing the primary flow physics.

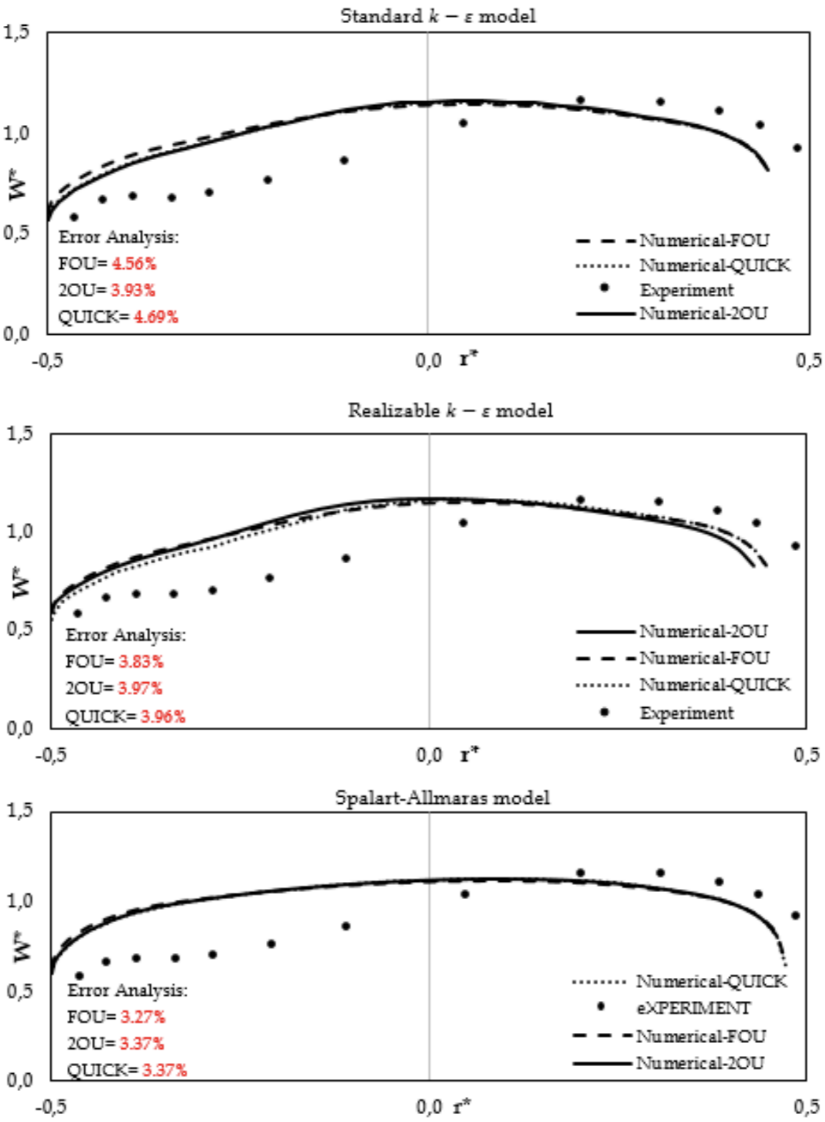


Figure 2.1: Streamwise velocity profiles in a 90° pipe bend ($R_c = 7D$, $Re_D \approx 34,000$, $De \approx 9,000$). The SA model performs relatively well, closely matching the experimental data and two-equation models for mild curvature. Reprinted from Apalowo [10].

However, as the curvature tightens and the Dean number increases, the limitations of the isotropic Boussinesq assumption become severe. Abuhatira et al. [1] evaluated the internal turbulent flow through a 90° pipe bend characterized by a tighter curvature ratio of $R_c = 2.0D$. Operating at a bulk Reynolds number of $Re_D = 60,000$, this geometry generated strong centrifugal forces and a corresponding Dean number of $De \approx 30,000$. Unlike extreme benchmark cases where all isotropic models fail uniformly, this configuration distinctly isolates the shortcomings of the Spalart-Allmaras formulation. At the critical bend exit ($X/D = 0$), the SA model drastically mischaracterized the near-wall velocity profiles, completely missing the flow physics close to the inner wall that higher-order two-equation models (such as $k-\epsilon$ and $k-\omega$) resolved with much greater accuracy. This confirms that the failure is not attributed to inertial scaling, but is strictly driven by the model's inability to compensate for strong anisotropic centrifugal effects.

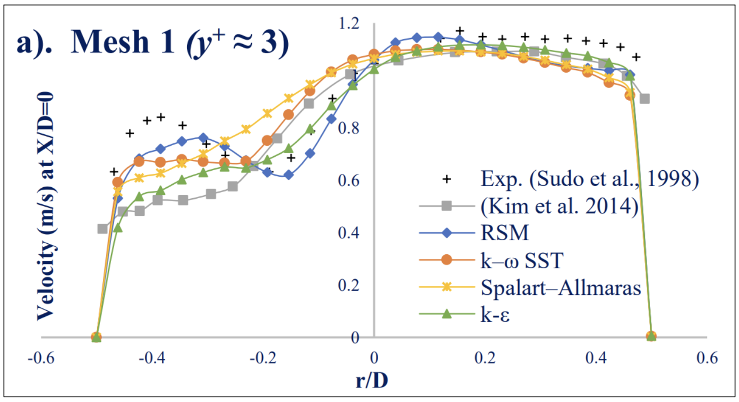


FIGURE 2.2: Turbulence model predictions at the 90° bend exit ($X/D = 0$, $R_c = 2.0D$, $Re_D = 60,000$, $De \approx 30,000$). Due to the strong curvature, the standard Spalart-Allmaras model fails to accurately capture the near-wall velocity profile compared to two-equation closures. Adapted from Abuhatira et al. [1].

While the literature clearly establishes that the SA model is viable at $De \approx 9,000$ and fails to capture inner-wall physics at $De \approx 30,000$, a bracketed or case-specific threshold between these states must be estimated to justify its use in topology optimization. Therefore, to better evaluate where the model begins to break down, rigorous benchmark studies were subsequently conducted within Ansys Fluent.

2.6 Benchmark Studies in Ansys Fluent

To investigate where the SA model's accuracy deteriorates, benchmark simulations were performed using **Ansys Fluent**. The study focused on two distinct, fully three-dimensional (3D) internal flow geometries: a 180° U-bend and a tighter 90° pipe bend. To ensure a standardized and rigorously tested numerical baseline, the computational meshes, fluid properties, and boundary conditions for these simulations were adopted directly from the official Ansys Fluid Dynamics Verification

Manual [9]. Specifically, the 90° bend corresponds to verification case VMFL030, while the 3D U-bend corresponds to case VMFL048. These cases evaluate industrial flow regimes ($Re_D > 40,000$) to ensure the turbulence model is rigorously verified against dominant inertial physics before being implemented into the custom FEniCS solver. To evaluate the accuracy of the one-equation model, the results were compared directly against the $k - \omega$ SST model, an industry standard for accurately predicting adverse pressure gradients and flow separation.

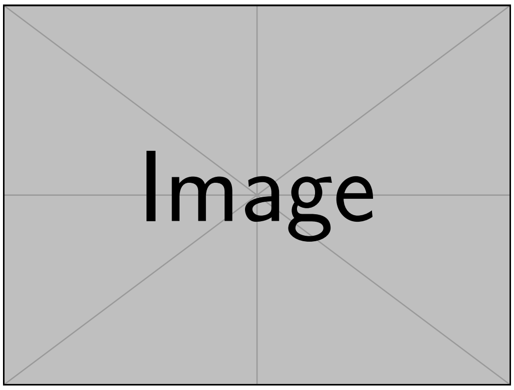
2.6.1 Mild Curvature: The U-Bend Case

The first benchmark evaluated a 3D U-bend geometry characterized by a moderate curvature ratio of $R_c \approx 4.46D$. Operating at a bulk Reynolds number of $Re_D \approx 44,700$, this geometry resulted in a Dean number of $De \approx 15,000$. Consistent with the findings of Apalowo for moderate bends, the SA model performed well under these conditions. As shown in Figure 2.3, the qualitative velocity contours of the SA model accurately replicate the bulk flow distribution of the $k - \omega$ SST model. Furthermore, the quantitative velocity profiles agree well with the experimental data from Takamasa and Kondo [43].

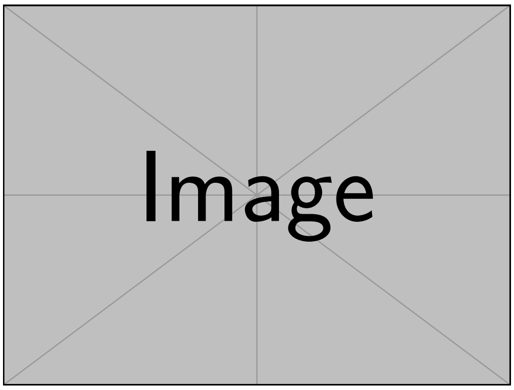
2.6.2 Strong Curvature: The 90° Bend Case

The second benchmark evaluated a tighter 90° pipe bend characterized by a severe curvature ratio of $R_c = 2.8D$. Despite operating at a very similar bulk Reynolds number of $Re_D \approx 43,000$ compared to the U-bend, this significantly tighter bend radius pushed the Dean number up to $De \approx 18,000$. Because the bulk flow speed and Reynolds number remain effectively constant between these two benchmark cases, this comparison successfully isolates the geometric effect of streamline curvature.

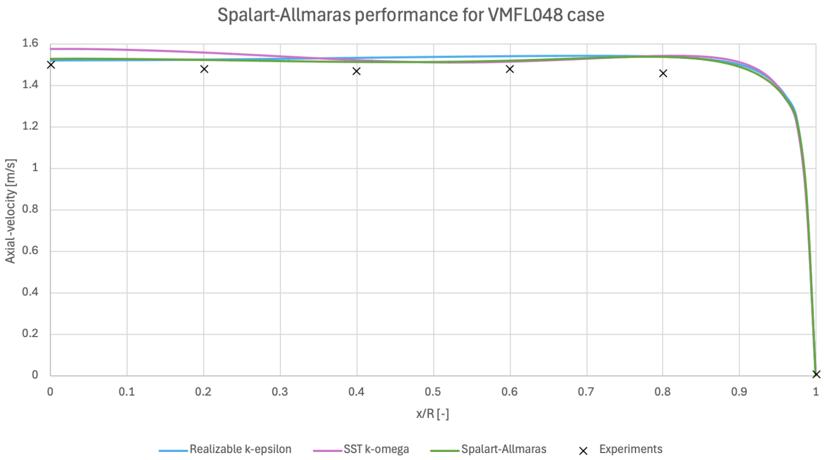
As shown in Figure 2.4, examining the quantitative velocity profile at the bend exit ($\theta = 90°$) clearly demonstrates the breakdown of the SA model. Driven primarily by the tighter bend radius and amplified anisotropic centrifugal effects, the SA model over-predicts the turbulent viscosity. This causes it to over-smooth the velocity profile, failing to capture the sharp velocity gradients near the inner wall ($y/D = 0$) when compared against both the $k - \omega$ SST prediction and the experimental data from Enayet et al. [19]. This failure is also visually apparent when comparing the side-by-side velocity contours.



(A) $k - \omega$ SST Contour

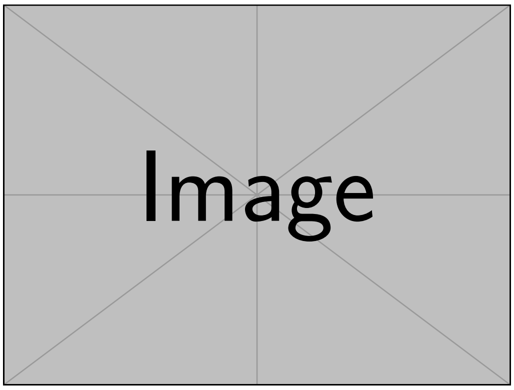


(B) Spalart-Allmaras Contour

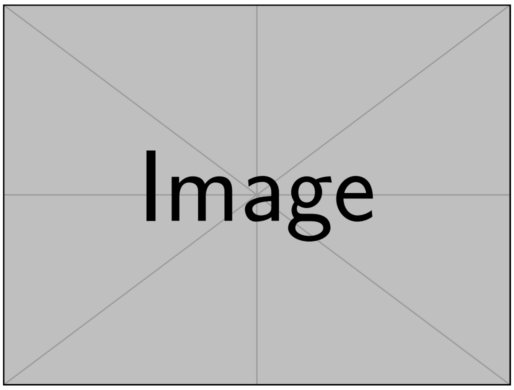


(C) Velocity Profiles

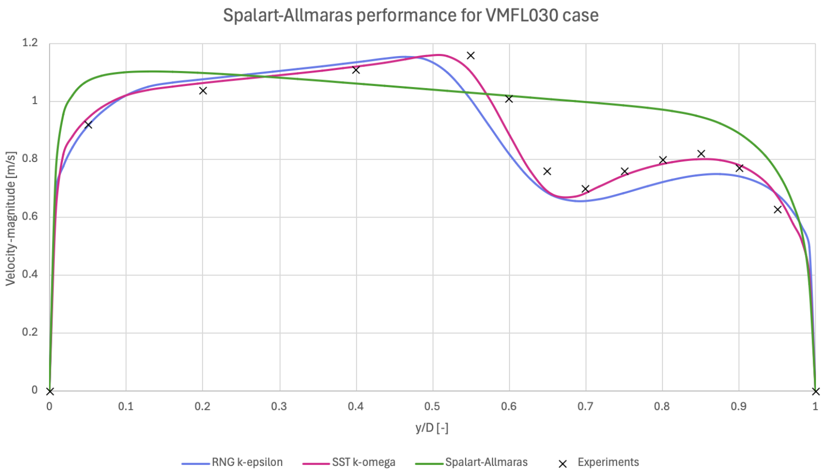
FIGURE 2.3: Flow predictions for a 180° U-bend ($R_c \approx 4.46D$, $Re_D \approx 44,700$, $De \approx 15,000$). The SA model accurately matches the higher-fidelity $k - \omega$ SST contours and directly aligns with the experimental data from Takamasa and Kondo [43].



(A) $k - \omega$ SST Contour



(B) Spalart-Allmaras Contour



(C) Velocity Profiles

Figure 2.4: Flow predictions at the 90° pipe bend exit ($R_c = 2.8D$, $Re_D \approx 43,000$, $De \approx 18,000$). The SA model over-smooths the velocity gradients near the inner wall, deviating significantly from both the $k - \omega$ SST model and the experimental data from Enayet et al. [19].

2.7 Conclusion

This chapter directly addresses the first research question regarding the predictive capabilities and physical limitations of the Spalart-Allmaras model. The theoretical breakdown and Ansys benchmark studies confirm that for internal geometries with moderate curvature (up to $De \approx 15,000$), the SA model performs accurately and successfully captures the primary flow physics. Furthermore, because it is a wall-resolved model, it avoids the numerical pitfalls of empirical wall functions, making it exceptionally well-suited for boundary-evolving topology optimization. However, its physical limitation lies in the isotropic Boussinesq assumption. When streamline curvature induces strong anisotropic secondary flows (heuristically observed to cause model breakdown in this case beyond a threshold of $De \approx 18,000$), the uncompensated SA model systematically overpredicts eddy viscosity and fails to capture flow separation. Consequently, while the SA model provides the necessary numerical stability to drive the optimization algorithms in FEniCS, the final topological designs must be critically verified. Any complex manifold generated by the SA optimizer must be subsequently evaluated in commercial software using a higher-fidelity two-equation model to confirm its true aerodynamic performance.

Chapter 3

Implementation and Verification of the Spalart-Allmaras Model in FEniCS

3.1 Introduction to FEniCS and the Finite Element Method

With the theoretical foundation of the Spalart-Allmaras (SA) model established in Chapter 2, this chapter details its numerical translation into a working computational solver. The numerical simulations and eventual topology optimization in this thesis are built upon FEniCS, an open-source computing platform for solving partial differential equations (PDEs). Unlike commercial computational fluid dynamics (CFD) software such as Ansys, which typically relies on the Finite Volume Method (FVM) and provides a "black box" graphical interface, FEniCS uses the Finite Element Method (FEM) and requires the user to explicitly define the mathematical formulation of the problem [30].

To solve a PDE in FEniCS, the governing equations (the "strong form") must be mathematically transformed into a "weak formulation" (or variational form). FEniCS utilizes the Unified Form Language (UFL) to express these weak forms in Python code that closely mirrors the mathematical notation [28]. This high-level mathematical scripting is particularly advantageous for topology optimization, as it allows for the automated derivation of the adjoint equations required for gradient-based optimization algorithms.

To ensure complete transparency and reproducibility, the entire custom codebase developed for this thesis is available in the public GitHub repository `TurbulentTO`. To maintain modularity, the repository is divided into two primary directories. The `TurbulenceModels` directory contains the standalone Spalart-Allmaras fluid dynamics engine and the verification benchmarks discussed throughout this chapter. Conversely, the `FluidTO` directory houses the optimization algorithms, penalty formulations, and objective functions that will be detailed in Chapters 4 and 5.

3.1.1 Derivation of a Weak Formulation

Before diving into the complex turbulence equations, it is instructive to demonstrate how a continuous PDE is translated into a discrete FEniCS formulation. To illustrate this methodology, consider the Poisson equation for pressure, which is a fundamental component of incompressible fluid solvers. The strong form of the Poisson equation is given by:

$$-\nabla^2 p = f \quad \text{in}\,\Omega \tag{3.1}$$

where p is the pressure field, f is a source term (derived from the velocity divergence), and Ω is the computational domain. To obtain the weak form, the equation is multiplied by a sufficiently smooth test function v and integrated over the entire domain Ω, following standard Galerkin finite element procedures [30, 28]:

$$-\int_\Omega (\nabla^2 p) v \, dx = \int_\Omega f v \, dx \tag{3.2}$$

Applying integration by parts (Green's first identity) to the left-hand side yields:

$$\int_\Omega \nabla p \cdot \nabla v \, dx - \int_{\partial\Omega} (\nabla p \cdot \mathbf{n}) v \, ds = \int_\Omega f v \, dx \tag{3.3}$$

where $\partial\Omega$ is the boundary of the domain and $\mathbf{n}$ is the outward-pointing normal vector. This integral formulation fundamentally lowers the continuity requirements of the solution, requiring only first-order derivatives to be square-integrable. Furthermore, Neumann boundary conditions (e.g., a prescribed pressure gradient at an outlet) can be naturally substituted directly into the boundary integral $\int_{\partial\Omega}(\nabla p \cdot \mathbf{n}) v \, ds$. This variational approach forms the foundation of all fluid and turbulence equations implemented in this thesis.

3.2 Numerical Solution Strategy for Fluid Dynamics

Solving the full Reynolds-Averaged Navier-Stokes (RANS) equations is numerically challenging due to the non-linear convective terms, the saddle-point nature of the coupled velocity-pressure system, and the intense cross-coupling with the turbulence model. To achieve a stable and efficient steady-state solution in FEniCS that seamlessly translates into the topology optimization framework, the governing equations are rigorously decoupled.

3.2.1 Decoupling the Physics: The Picard Iteration Scheme

Because the momentum equation relies on the turbulent eddy viscosity (ν_t), and the turbulence transport equation relies on the convective velocity ($\mathbf{u}$), solving the entire fluid-turbulence system simultaneously in a massive monolithic matrix frequently leads to ill-conditioning and solver divergence.

To resolve this, the forward physics are decoupled using a **Picard iteration scheme** (successive substitution), a robust approach for segregated RANS solvers

[21, 46]. During each Picard step, the flow solver advances using the eddy viscosity from the previous iteration. The newly updated velocity field is then passed to the Spalart-Allmaras solver to calculate a new eddy viscosity field. This alternating sequence forms the outer loop of the solver, iteratively updating the decoupled equations until the fully coupled physical system reaches a steady-state equilibrium.

3.2.2 Incremental Pressure Correction Scheme (IPCS)

Within the Picard loop, the Navier-Stokes equations themselves must be solved. Rather than solving for velocity and pressure simultaneously, this implementation utilizes an Incremental Pressure Correction Scheme (IPCS). This algorithmic splitting is fundamentally based on the fractional-step projection methods originally pioneered by Chorin [16] and Temam [44]. To nurse the non-linear convective momentum terms to a stable steady state, a pseudo-time stepping approach (Δt) is employed strictly for the flow variables. The IPCS algorithm splits the fluid equations into three sequential steps per pseudo-time step:

1. **Tentative velocity step:** The momentum equation is solved using the pressure field from the previous iteration (p^n) to compute an intermediate, non-divergence-free velocity field ($\mathbf{u}^*$). Because this is a RANS framework, the diffusion term utilizes the effective kinematic viscosity, ($\nu + \nu_t$). Treating the convective term explicitly and the diffusive term implicitly, the semi-discrete equation is:

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + (\mathbf{u}^n \cdot \nabla)\mathbf{u}^* - \nabla \cdot ((\nu + \nu_t^n)\nabla \mathbf{u}^*) = -\nabla p^n \tag{3.4}$$

2. **Pressure correction step:** A Poisson equation is solved to find the pressure variation that enforces the continuity constraint ($\nabla \cdot \mathbf{u}^{n+1} = 0$). The equation for the new pressure field (p^{n+1}) is derived as:

$$\nabla^2 p^{n+1} = \nabla^2 p^n + \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^* \tag{3.5}$$

3. **Velocity update step:** The tentative velocity is strictly corrected using the gradient of the pressure difference to yield a mass-conserving velocity field ($\mathbf{u}^{n+1}$) for the current iteration:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla (p^{n+1} - p^n) \tag{3.6}$$

The simulation is deemed fully converged to a steady state when the L_2 norm of the residual difference between successive Picard iterations falls below a defined tolerance limit (ϵ_{tol}) for both the fluid and turbulence fields.

3.3 Detailed Formulation of the Spalart-Allmaras Model

As concluded in Chapter 2, the Spalart-Allmaras (SA) one-equation model was selected for its balance of physical accuracy and numerical stability. Unlike the transient flow equations, the SA transport equation is solved in its purely steady-state form within the Picard loop. The steady transport equation for the continuous nonlinear working variable $\tilde{\nu}$ is given by:

$$\mathbf{u} \cdot \nabla\tilde{\nu} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y}\right)^2 + \frac{1}{\sigma}\left[\nabla \cdot ((\nu + \tilde{\nu})\nabla\tilde{\nu}) + c_{b2}(\nabla\tilde{\nu})^2\right] \tag{3.7}$$

where ν is the molecular kinematic viscosity, $\sigma = 2/3$ is the turbulent Prandtl number, c_{b1} and c_{w1} are calibrated closure constants, $\tilde{S}$ is a modified vorticity magnitude, f_w is the wall-damping function, and y is the physical distance to the nearest solid boundary [40].

3.3.1 Closure Terms and Calibration Constants

The Spalart-Allmaras model relies on several interconnected closure functions to accurately capture boundary layer physics. The production term relies on a modified vorticity magnitude, $\tilde{S}$, which prevents the destruction of turbulence in regions where the vorticity goes to zero, while maintaining correct scaling near the wall [40]. It is defined as:

$$\tilde{S} = S + \frac{\tilde{\nu}}{\kappa^2 y^2} f_{v2} \tag{3.8}$$

where $S = \sqrt{2\Omega_{ij}\Omega_{ij}}$ is the scalar magnitude of the vorticity tensor, κ is the von Kármán constant, and y is the distance to the nearest wall. The function f_{v2} is a near-wall modifier:

$$f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad \text{where} \quad \chi = \frac{\tilde{\nu}}{\nu} \quad \text{and} \quad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3} \tag{3.9}$$

Similarly, the wall destruction term is modulated by the non-dimensional function f_w, which accelerates the decay of turbulent viscosity inside the viscous sublayer:

$$f_w = g\left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6}\right)^{1/6} \tag{3.10}$$

where the intermediate variables g and r are evaluated as:

$$g = r + c_{w2}(r^6 - r) \quad \text{and} \quad r = \min\left(\frac{\tilde{\nu}}{\tilde{S}\kappa^2 y^2}, 10\right) \tag{3.11}$$

The upper bound of 10 on r ensures numerical stability during solver initialization when $\tilde{S}$ may momentarily approach zero.

It should be explicitly noted that the original formulation by Spalart and Allmaras [40] includes additional geometric trip functions (f_{t1} and f_{t2}) designed to manually trigger the transition from laminar to turbulent flow at specific user-defined coordinates. However, following the topology optimization methodology established by Yoon [49], these transition functions are deliberately omitted from this thesis. Because the topological boundaries are unknown *a priori* and evolve during optimization, tracking discrete trip points is mathematically unfeasible; therefore, the model assumes a fully turbulent flow regime.

To close the system, the standard empirical calibration constants formulated by Spalart and Allmaras are utilized throughout the FEniCS implementation [37]. Notably, the von Kármán constant is set to 0.41 to perfectly align with the standard OpenFOAM and generalized industrial implementations[cite: 3]:

$$
\begin{aligned}
c_{b1} &= 0.1355 & \sigma &= 2/3 & c_{b2} &= 0.622 \\
\kappa &= 0.41 & c_{w2} &= 0.3 & c_{w3} &= 2.0 \\
c_{v1} &= 7.1 & c_{w1} &= \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma} & &
\end{aligned}
$$

3.3.2 Wall-Distance Calculation via the Relaxed Eikonal Equation

A major computational challenge in the SA model is the calculation of the wall distance y. In a standard static CFD simulation, y is purely geometric and can be pre-computed. However, to ensure this verification solver translates seamlessly into the topology optimization code (where solid boundaries evolve continuously), the implementation abandons discrete geometric searches in favor of a PDE-based approach.

The true geometric wall distance satisfies the standard Eikonal equation, $|\nabla y| = 1$. However, this formulation contains singularities at the medial axes of the domain. To guarantee continuous differentiability—a requirement for later adjoint derivations—a relaxed Eikonal equation proposed by Yoon [49] is solved:

$$\nabla G \cdot \nabla G + \sigma_w G(\nabla \cdot \nabla G) = (1 + 2\sigma_w)G^4 \tag{3.12}$$

Here, G is an intermediate, non-dimensional field variable. The parameter σ_w is a relaxation constant (typically set to 0.1) that controls the smoothness of the field. The PDE is subjected to a Dirichlet boundary condition at the physical walls, $G = G_0$, where G_0 is an arbitrarily large constant (e.g., $G_0 = 20.0$). Once the field G is solved over the domain, the physical wall distance y is recovered algebraically at every node using the inverse relationship:

$$y = \frac{1}{G} - \frac{1}{G_0} \tag{3.13}$$

This system is implemented in the core solver suite within the `Utilities.py` script. As shown in Listing 3.1, once the non-linear variational problem for G is solved, the physical distance y is recovered and strictly bounded from below to prevent numerical singularities.

LISTING 3.1: Implementation of Yoon's relaxed Eikonal equation for wall-distance calculation, extracted from `Utilities.py`.

```
# Weak form corresponding to Yoon 2016 Eq. (19):
F = (
(Constant(1.0) - sigma_w_const) * inner(grad(G), grad(G)) * z
    * dx
- sigma_w_const * G * inner(grad(G), grad(z)) * dx
- (Constant(1.0) + Constant(2.0) * sigma_w_const) * G**4 * z *
     dx
)
problem = NonlinearVariationalProblem(F, G, bcs=wall_bcs, J=
    derivative(F, G))
solver = NonlinearVariationalSolver(problem)
solver.solve()

# Bound G to prevent division by zero
G.vector()[:] = np.maximum(G.vector()[:], float(g_floor))

# Recover physical wall distance (y)
y = project(Constant(1.0) / G - Constant(1.0) / g0_const,
    Space)
return bound_from_bellow(y, 0.0)
```

3.4 Variational Formulation and FEniCS Implementation

The standard Galerkin weak formulation is constructed by multiplying the steady transport Equation 3.7 by a test function ξ and applying integration by parts to the diffusive terms. Because the equation is highly non-linear, robust convergence is dependent on the stability of the external velocity field provided by the Picard outer loop.

To manage this non-linearity, the equation is linearized within the Picard iteration scheme by freezing certain parameters using the most recent value of the working variable, $\tilde{\nu}^0$. Specifically, the turbulence production and cross-diffusion terms are treated as explicit sources evaluated using $\tilde{\nu}^0$[cite: 3]. Conversely, the wall destruction term is linearized into an implicit reaction sink by scaling the continuous $\tilde{\nu}$ against the frozen variable, taking the form $c_{w1} f_w(\tilde{\nu}^0/y^2)\tilde{\nu}$[cite: 3].

The explicit translation of this mathematical formulation is handled by the `SpalartAllmarasSteadyState` class within the `TurbulenceModel_SpalartAllmaras.py` script[cite: 3]. As demonstrated in Listing 3.2, the Unified Form Language (UFL) mirrors the analytical weak form utilizing the most recent convective velocity field (`external_velocity`) passed from the IPCS flow solver.

LISTING 3.2: FEniCS implementation of the steady-state SA standard Galerkin weak formulation. Python dictionary lookups have been simplified for readability.

```
def construct_forms(self, external_velocity):
self._construct_turbulent_quantities(external_velocity)
sigma = Constant(2.0 / 3.0)
```

```
# Steady-state SA transport equation weak form
FNT = (
dot(external_velocity, nabla_grad(self._nu_tilde)) * self._xi
    * self._dx
+ inner((self._nu + self._nu_tilde0) / sigma * grad(self.
    _nu_tilde), grad(self._xi)) * self._dx
+ self._react_nt * self._nu_tilde * self._xi * self._dx
- self._source_nt * self._xi * self._dx
)

self._a_nt = lhs(FNT)
self._l_nt = rhs(FNT)
```

3.5 General Solver Configuration and Numerical Parameters

Before detailing the specific verification benchmarks, it is necessary to establish the general numerical configuration of the FEniCS solver. Both the channel flow and the U-bend verifications were simulated using the segregated algorithmic structure detailed earlier: an outer Picard fixed-point loop couples the fluid momentum equations to the Spalart-Allmaras transport equation, while an inner IPCS resolves the velocity-pressure subsystem via pseudo-time stepping.

Because the U-bend domain introduces severe geometric curvature, strong convective transport, and adverse pressure gradients, it represents a significantly more demanding numerical challenge than a fully developed, unidirectional channel flow. Consequently, the U-bend simulations necessitated specific numerical controls to maintain stability, including a heavily refined IPCS pseudo-time step ($\Delta t = 1.0 \times 10^{-4}$ s), a maximum of 400 coupled Picard iterations, up to 800 IPCS steps per Picard update, a pressure relaxation factor of 1.0, and tighter under-relaxation of the turbulence field updates ($\alpha_{\tilde{\nu}} = 0.01$)[cite: 2]. To ensure reproducibility, the comprehensive matrices detailing all physical boundary conditions, geometric dimensions, and exact numerical solver parameters for both benchmarks have been compiled and are documented extensively in Sections A.1 and A.2 of Appendix A (specifically in Tables A.2 and A.6).

3.6 Verification Benchmark 1: Channel Flow

To verify the fundamental correctness of the custom fluid and turbulence solvers, the code was first tested on a standard 2D turbulent channel flow. To cleanly isolate and verify turbulence production, wall damping, and Picard coupling without the artifacts of a developing boundary layer, this benchmark was formulated as a pressure-driven periodic channel rather than a conventional inlet-outlet duct.

The velocity field ($\mathbf{u}$) and Spalart-Allmaras working variable ($\tilde{\nu}$) were constrained periodically between the streamwise boundaries, while the top and bottom boundaries were treated as standard no-slip walls ($\mathbf{u} = 0, \tilde{\nu} = 0$). Flow was driven entirely by a fixed pressure drop from the upstream plane ($p = 2$ Pa) to the downstream plane ($p = 0$ Pa). Based on a reference bulk velocity of $U_{\text{ref}} = 18.5$ m/s and utilizing the full channel height ($H = 2.0$ m) as the characteristic length scale, the solver was targeted to resolve a fully turbulent flow regime at a channel Reynolds number of $Re_H \approx 20,300$. By explicitly defining the Reynolds number based on the channel height rather than the mathematical hydraulic diameter, this formulation directly aligns with academic conventions for fundamental 2D channel flow verification. The complete definition of the geometric, physical, and boundary-condition parameters can be found in Table A.1 of Appendix A.

3.6.1 Mesh Resolution Requirements

Because the SA formulation is a "low-Reynolds" model that does not utilize empirical wall functions, it requires mathematical integration directly through the viscous sublayer. To satisfy this strict boundary layer requirement, a custom structured inflation mesh was generated for the channel flow. Because the domain is a simple periodic rectangle, this highly anisotropic grid was constructed programmatically utilizing NumPy arrays and converted into a FEniCS-compatible format via the `meshio` library. Based on the target bulk Reynolds number, this approach enforced a physical first-cell height of approximately 1.7×10^{-3} meters, aiming for a non-dimensional wall distance of $y^+ \approx 1$ across the entire domain.

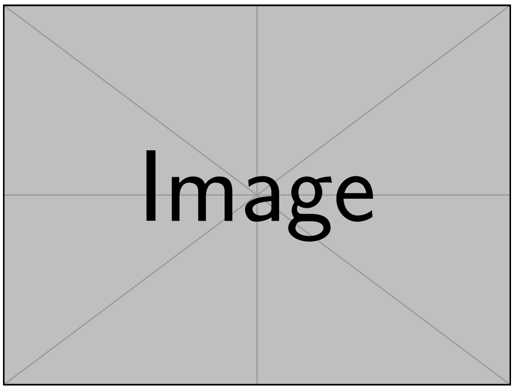
Furthermore, to verify that the numerical results were not heavily influenced by spatial discretization, a formal grid independence study was conducted. The full results of this spatial verification are documented in Appendix A.1.3. Specifically, the geometric properties of the mesh hierarchy are detailed in Table A.3, and visual confirmation of asymptotic convergence is provided by the velocity profiles in Figure A.1.

3.6.2 FEniCS vs. Ansys Comparison

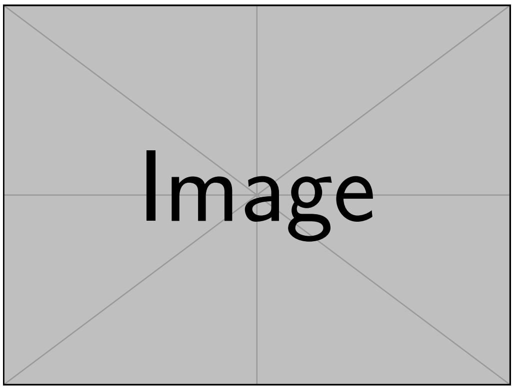
Prior to this comparative analysis, the Ansys baseline itself was verified for grid independence to ensure a robust standard (see Table A.4 and Figure A.2 in Appendix A). The channel flow was then simulated in FEniCS using the decoupled Picard iteration method and compared against the Ansys Fluent baseline.

As shown in Figure 3.1, the qualitative velocity contours of the custom FEniCS implementation show good agreement with the Ansys Fluent baseline. Furthermore, the quantitative line plots confirm that the velocity profiles generated by the custom FEniCS model closely track the Ansys reference data. The custom solver resolves the logarithmic region of the turbulent boundary layer and predicts the turbulent diffusion toward the channel center. To provide a rigorous, objective comparison, the quantitative errors and macroscopic metrics are summarized in Table 3.1.

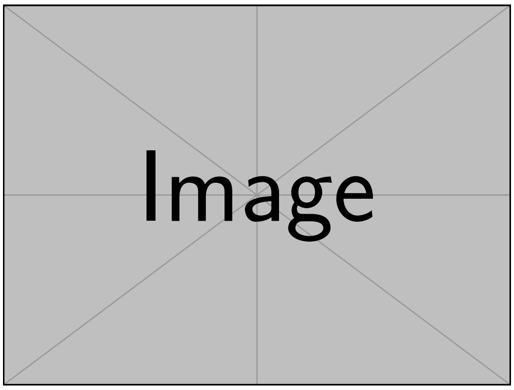
This strong alignment supports the verification that the core Navier-Stokes solver, the relaxed Eikonal field, and the SA production/destruction terms are functioning as intended before introducing the geometric complexities of the U-bend.



(a) Ansys Fluent SA Contour



(b) FEniCS SA Contour



(c) Velocity Profiles

Figure 3.1: Verification of the 2D Channel Flow ($Re_H \approx 20,300$). Comparison of the qualitative velocity contours between the Ansys Fluent baseline (a) and the custom FEniCS implementation (b), alongside the quantitative velocity profiles extracted across the channel (c).

Table 3.1: Quantitative comparison of flow metrics and extracted errors for the 2D channel benchmark.

Metric	**Ansys Fluent**	**Custom FEniCS**	**Relative Error / Difference**
Bulk Velocity (U_{bulk}) [m/s]	[TBD]	[TBD]	[TBD]%
Friction Coefficient (C_f)	[TBD]	[TBD]	[TBD]%
Max Pointwise Velocity Error	–	–	[TBD]%
Relative L_2 Velocity Error	–	–	[TBD]%

3.7 Verification Benchmark 2: U-Bend Geometry

Following the channel flow verification, the solver was tested on the highly convective separating flows of the U-bend geometry evaluated in Chapter 2.

As discussed in Appendix A.2.3, the verification performed here utilized a modified computational domain to ensure efficiency. First, the domain was reduced to a 2D geometry corresponding directly to the symmetry mid-plane of the full 3D case. Since the FEniCS domain is mathematically a 2D planar approximation, it inherently cannot resolve true 3D Dean secondary vortices. Instead, this benchmark serves to validate the 2D symmetry-plane approximation and the solver's ability to capture curvature-induced separation and momentum redistribution. Second, to reduce unnecessary computational overhead during these verification sweeps, the extended straight inlet and outlet sections (1.555 m) of the standard benchmark were truncated to $30D$ (0.840 m). The mathematical justification for this truncation is plotted in Figure A.3 of Appendix A.2.3.

To drive the flow, a uniform prescribed velocity ($\mathbf{u}_{in} = 1.42$ m/s) was imposed at the inlet alongside a fully turbulent boundary value ($\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m^2/s), while a zero relative static pressure ($p = 0$ Pa) was assigned to the outlet. While the domain is 2D, the 3D physical pipe diameter ($D = 0.028$ m) and geometric radius of curvature ($R_c = 0.125$ m) were retained as the characteristic length scales. Conserving these parameters ensures that the computed Reynolds number ($Re_D \approx 44,700$) and Dean number ($De \approx 15,000$) match the 3D Ansys reference case. For an overview of the physical setup and boundary conditions applied to this geometry, the reader is referred to Table A.5 in Appendix A.

3.7.1 Mesh Resolution Requirements

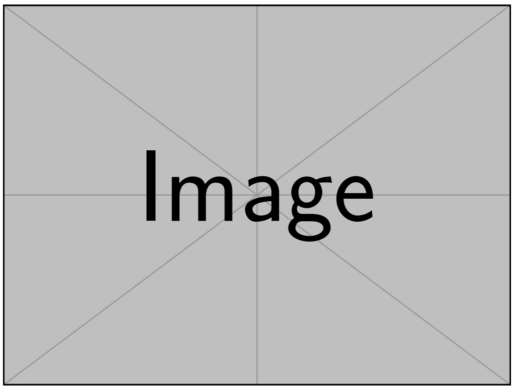
Because this formulation utilizes a standard Galerkin weak form without spatial upwind stabilization, numerical stability in highly convective flows relies on adequate grid resolution.

The U-bend geometry features curved inner and outer arcs, complex corner transitions, and anisotropic boundary-layer inflation around non-orthogonal walls. To manage this geometric complexity while satisfying the $y^+ \approx 1$ resolution requirement, a custom mesh was generated using **Gmsh** [23]. The first element height adjacent to the wall was controlled to 1.20×10^{-5} meters across the entire domain. As documented in Appendix A.2.4, independence studies were conducted. The inflation parameters and element densities of the Gmsh hierarchy are listed in Table A.7, and the extracted velocity profiles confirming grid convergence at the bend exit are illustrated in Figure A.4.

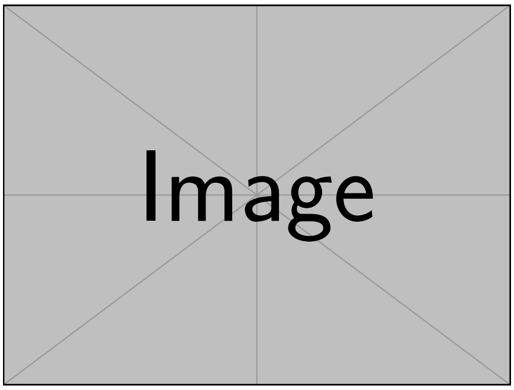
3.7.2 FEniCS vs. Ansys Comparison

Having validated the spatial resolution of the FEniCS mesh, an equivalent grid independence study was simultaneously executed for the commercial Ansys Fluent baseline (detailed in Table A.8 and Figure A.5 of Appendix A.2.4). With both solvers operating on spatially converged grids, an independent Ansys Fluent simulation was executed on the shortened 2D domain.

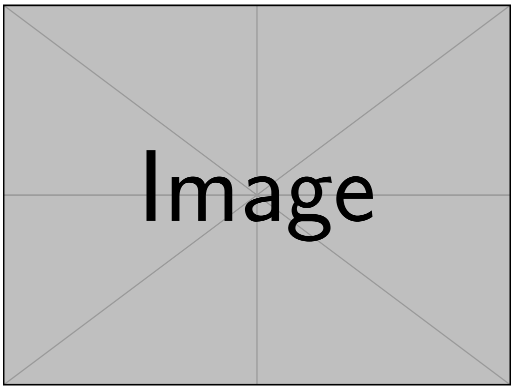
As shown in Figure 3.2, the global flow fields and flow separation regions observed in the qualitative velocity contours demonstrate visual agreement. To provide a more robust assessment, velocity data was extracted along a cross-sectional line probe



(a) Ansys Fluent SA Contour



(b) FEniCS SA Contour



(c) Velocity Profiles

Figure 3.2: Verification of the 2D U-Bend Flow ($Re_D \approx 44,700$, $De \approx 15,000$). Comparison of the qualitative velocity contours between the Ansys Fluent baseline (a) and the custom FEniCS implementation (b), showing separation characteristics, alongside the quantitative velocity profiles extracted at the bend exit (c).

positioned directly at the bend exit. The quantitative velocity profile computed by the custom FEniCS implementation traces the commercial Ansys results closely, adequately capturing the boundary layer growth and the momentum shift toward the outer wall.

Table 3.2 summarizes the quantitative error margins and the macroscopic pressure drop (Δp) across the 2D U-bend domain for both solvers.

Table 3.2: Quantitative comparison of flow metrics and extracted errors for the 2D U-bend benchmark.

Metric	**Ansys Fluent**	**Custom FEniCS**	**Relative Error / Difference**
Pressure Drop (Δp) [Pa]	[TBD]	[TBD]	[TBD]%
Max Pointwise Velocity Error	–	–	[TBD]%
Relative L_2 Velocity Error	–	–	[TBD]%

This general agreement in both localized velocity gradients and global energy dissipation indicates that the framework provides a credible foundation for the pressure-drop minimization objectives in the subsequent topology optimization campaigns.

3.8 Conclusion

This chapter has detailed the mathematical formulation, stabilization strategy, and verification of the steady-state Spalart-Allmaras model within the FEniCS framework. By evaluating the solver against Ansys Fluent benchmarks, this implementation addresses the second research question of this thesis: achieving numerical stability and establishing reasonable quantitative agreement in a custom finite element solver to support topology optimization routines.

First, numerical stability was achieved without the need for complex mathematical stabilization artifacts. By decoupling the Navier-Stokes and Spalart-Allmaras equations via a Picard iteration scheme, and utilizing an Incremental Pressure Correction Scheme (IPCS) to manage the convective fluid terms, the highly non-linear system was driven to equilibrium. Supported by high-density mesh resolution and Yoon's relaxed Eikonal equation, this standard Galerkin formulation effectively mitigated numerical oscillations.

Second, quantitative agreement was assessed. The solver produced velocity profiles that showed good agreement with the commercial Ansys Fluent baseline in simple periodic channel flows ($Re_H \approx 20,300$) and adequately captured the 2D boundary layer features in a curved U-bend ($Re_D \approx 44,700$, $De \approx 15,000$). While approximations inherent to 2D domains limit the direct translation to true 3D Dean vortices, this decoupled Picard FEniCS solver demonstrates sufficient accuracy in capturing curvature-induced momentum shifts. This suggests that the implementation is suitable for driving the continuous adjoint topology optimization campaigns presented in Chapters 4 and 5.

Chapter 4

Topology Optimization for Turbulent Flows: Methodology

4.1 Introduction

Building upon the verified Spalart-Allmaras (SA) fluid dynamics solver developed in Chapter 3, this chapter introduces the mathematical framework required to perform Topology Optimization (TO) in the turbulent regime. The core objective of this methodology is to algorithmically determine the optimal distribution of fluid channels within a given design domain to minimize energy dissipation.

This chapter details the Fictitious Domain Method and the Brinkman penalization technique used to conceptualize solid boundaries within a continuous fluid mesh. It explicitly outlines how the standard governing equations for fluid momentum, turbulent transport, and wall distance must be mathematically penalized to accommodate the evolving topology.

To solve this optimization problem in a computationally tractable manner, continuous adjoint sensitivity analysis is required. This chapter presents two distinct adjoint pipelines. First, the "frozen turbulence" (Constant Eddy Viscosity) assumption is established and rigorously justified as the stable baseline optimization engine adopted in this thesis. Subsequently, an advanced, monolithic velocity-pressure-SA adjoint formulation, termed the "semi-frozen" adjoint, is introduced to capture the highly non-linear turbulence production sensitivities. This section details both its physical advantages and its algorithmic limitations.

Following the derivation of these sensitivities, the continuous optimization problem is formally defined, establishing the power dissipation objective function and the strict fluid volume constraints over explicitly defined design regions. Finally, the mathematical techniques for active-domain density filtering and Heaviside projection are detailed alongside their FEniCS unified form language (UFL) implementation to ensure the generation of manufacturable designs. All algorithms, penalty formulations, and execution scripts, most notably `TurbulentTO_Frozen.py` and `TurbulentTO_SemiFrozen.py`, are housed within the `FluidTO` directory of the public repository.

4.2 The Fictitious Domain Method and Brinkman Penalization

In standard shape optimization, the solid-fluid boundary is explicitly defined by the mesh, and the nodes are physically moved to improve the design. In density-based topology optimization, however, the computational domain (Ω) remains strictly fixed. To create solid walls and fluid channels within this fixed grid, the Fictitious Domain Method is employed.

Originally applied to Stokes flow by Borrvall and Petersson and extended to the Navier-Stokes equations by Gersborg-Hansen et al., this method assigns a continuous design variable to the spatial coordinates in the domain. In this specific implementation convention, a physical topology value of $\bar{\gamma} = 1$ represents a pure **fluid** channel, while $\bar{\gamma} = 0$ represents a **solid**, impermeable structure. To force the fluid velocity to zero inside the solid regions without mathematically removing the finite elements, the fluid is treated as a fictitious porous medium.

This flow resistance is quantified by an inverse permeability term, or Brinkman penalty function, denoted as $\alpha(\bar{\gamma})$. To ensure the optimization algorithm (which relies on continuous gradients) can transition smoothly between fluid and solid states, α must be a continuous interpolation function. This solver utilizes a convex Rational Approximation of Material Properties (RAMP) interpolation scaled between a maximum solid penalty (α_{solid}) and a minimum fluid penalty (α_{fluid}):

$$\alpha(\bar{\gamma}) = \alpha_{solid} + (\alpha_{fluid} - \alpha_{solid}) \frac{\bar{\gamma}(1+q)}{\bar{\gamma}+q} \tag{4.1}$$

where α_{solid} is a sufficiently large penalty factor that completely stagnates the flow when $\bar{\gamma} = 0$, and q is a tuning parameter that controls the convexity of the penalization curve. Unlike traditional structural optimization which relies on the SIMP power-law, fluid optimization generally avoids SIMP for momentum because its derivative vanishes at $\bar{\gamma} = 0$. By utilizing the RAMP formulation, a strictly non-zero gradient is guaranteed across the entire design domain.

4.2.1 The Density Chain and Domain Bounding

While conceptually represented by the physical topology field in the governing equations, the computational implementation dictates that the layout is managed through a distinct chain of density representations to manage design and non-design domains. To explicitly avoid mathematical confusion with the fluid mass density (ρ) utilized in the Reynolds-Averaged Navier-Stokes equations, this thesis exclusively utilizes the γ notation for all topological variables. However, these variables correspond directly to the exact implementation code variables as follows:

- **Raw design density** ($\gamma \equiv \rho_{\text{raw}}$): The baseline parameter manipulated by the optimizer.
- **Filtered density** ($\tilde{\gamma} \equiv \rho_f$): The spatially smoothed design field.

- **Projected effective density** ($\bar{\gamma} \equiv \rho_{\text{eff}}$): The bounded, physical topology field evaluated by the governing equations.

The raw design density, γ, is defined as a Discontinuous Galerkin (DG0) function over the entire mesh. The optimization algorithm strictly manipulates the degrees of freedom corresponding to the actively optimized design space (`ActiveDV`), while passive (non-design) cells are subsequently restored to their prescribed bounding values.

This raw variable is subjected to a spatial PDE filter to yield a smoothed field $\tilde{\gamma}$. To recover sharp boundaries and strictly preserve required non-design regions, the filtered density is then projected and mapped into the bounded effective density ($\bar{\gamma}$):

$$\bar{\gamma} = \gamma_{\text{lower}} + (\gamma_{\text{upper}} - \gamma_{\text{lower}})\Pi_{\beta}(\tilde{\gamma}) \tag{4.2}$$

where Π_{β} is the projection operator, and γ_{lower} and γ_{upper} define the absolute permissible bounds for any given cell. This rigorous bounding guarantees that predefined features, such as non-design fluid inlet extensions ($\gamma_{\text{lower}} = \gamma_{\text{upper}} = 1$) or explicitly constrained structural walls, remain rigidly fixed without interfering with the mathematical behavior of the active optimization space.

4.3 Penalization of the Governing Equations

To effectively optimize turbulent manifolds, the respective penalties must be rigorously integrated into the turbulence model and the wall-distance calculations to ensure the physics remain consistent as the solid boundaries evolve. All physical penalties strictly utilize the projected effective density, $\bar{\gamma}$.

4.3.1 Modified Reynolds-Averaged Navier-Stokes Equations

The primary application of the Brinkman penalty is the modification of the steady-state, incompressible RANS momentum equation. The RAMP-interpolated inverse permeability $\alpha(\bar{\gamma})$ is added as a linear sink term (Darcy friction) to the momentum conservation equation:

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot \left[(\mu + \mu_t)\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right)\right] - \alpha(\bar{\gamma})\mathbf{u} \tag{4.3}$$

It is critical to note the retention of the full symmetric rate-of-strain tensor. Here, ρ strictly denotes the fluid mass density. Because the turbulent eddy viscosity (μ_t) varies drastically across the spatial domain, the effective fluid viscosity is no longer constant, and the transposed term exerts a non-zero physical influence on the momentum balance. In regions where the optimizer places solid material ($\bar{\gamma} \to 0$), the enormous magnitude of $\alpha_{solid}\mathbf{u}$ dominates all inertial and viscous forces, forcing the local velocity mathematically to zero.

4.3.2 Modified Spalart-Allmaras Transport Equation

As the Fictitious Domain Method forces velocity to zero in the solid regions, the turbulence generation inside those regions should theoretically cease. However, due to numerical diffusion, turbulent eddy viscosity ($\tilde{\nu}$) can inadvertently bleed into the solid regions. This unphysical behavior destabilizes the solver and heavily distorts the adjoint sensitivity analysis.

To strictly enforce a laminar, zero-turbulence state inside the solid structures, a corresponding penalty sink term is added to the Spalart-Allmaras transport equation:

$$\mathbf{u}\cdot\nabla\tilde{\nu} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1}f_w\left(\frac{\tilde{\nu}}{y}\right)^2 + \frac{1}{\sigma}\left[\nabla\cdot((\nu+\tilde{\nu})\nabla\tilde{\nu}) + c_{b2}(\nabla\tilde{\nu})^2\right] - \alpha_{\tilde{\nu}}[\max(1-\bar{\gamma},0)]^{N_{\tilde{\nu}}}\tilde{\nu} \tag{4.4}$$

To achieve this isolation numerically, the implementation intentionally diverges from the RAMP interpolation used for momentum, leveraging a positive-part power-law penalty instead. The penalization magnitude is determined by an active-solid indicator, $[\max(1-\bar{\gamma},0)]^{N_{\tilde{\nu}}}$, where the exponent $N_{\tilde{\nu}}$ strictly scales the sink term exclusively inside the designated solid bounds. By relying on this positive-part formulation and staged penalty amplitudes ($\alpha_{\tilde{\nu}}$) governed by the continuation schedule, the solver strongly suppresses the penalty in near-fluid regions while overwhelmingly destroying spurious turbulence generation in the solids.

4.3.3 Modified Relaxed Eikonal Equation

The Spalart-Allmaras model is highly sensitive to the wall distance, y. While differential equations for wall distance computation are well-established, topology optimization introduces a unique challenge. The location of the wall is not a static mesh boundary, but rather the dynamically evolving $\bar{\gamma} = 0$ contour. Therefore, the relaxed Eikonal equation must be penalized to dynamically recognize the fictitious solid as a physical boundary:

$$\nabla G\cdot\nabla G + \sigma_w G(\nabla\cdot\nabla G) = (1+2\sigma_w)G^4 + \alpha_G I_s^{N_G}(G - G_0) \tag{4.5}$$

Consistent with the SA transport modifications, the Eikonal equation utilizes a staged multiplier α_G paired with a power-law solid indicator. However, to provide further geometric control, the wall distance penalty incorporates an explicit solid threshold parameter (θ_s). The thresholded solid indicator is formulated as $I_s = \max((\theta_s - \bar{\gamma})/\theta_s, 0)$, which effectively tightens the geometric restrictions (forcing $G \to G_0$) only where the physical topology field drops below the designated solid threshold. If no explicit solid threshold below 1.0 is configured, this indicator naturally defaults to $I_s = \max(1-\bar{\gamma}, 0)$. This allows the algebraic recovery function $y = 1/G - 1/G_0$ to correctly evaluate to an absolute wall distance of $y = 0$ inside the generated structural material.

4.4 Boundary Conditions and Turbulence Initialization

To stabilize the topology optimization and ensure physical accuracy, boundary constraints are strictly managed across both fluid and turbulence regimes. Non-design fluid buffer regions are strictly mandated at inlets and outlets to isolate the active optimization domain from artificial numerical shear. The framework robustly handles boundary interactions by incorporating pressure outlets coupled with optional velocity-component constraints, as well as pressure pin controls to fully constrain the discrete PDE problem.

At the domain inlets, establishing accurate turbulent boundary conditions natively bridges physical intent with numerical stability. The prescribed turbulence intensity and the associated turbulence length scale are systematically converted into an intermediate eddy-viscosity ratio. This ratio directly calculates the requisite boundary value for the Spalart-Allmaras working variable $\tilde{\nu}$. Throughout the computation, to protect the solver from chaotic geometric fluctuations during intermediate design cycles, rigorous numerical floors, ceilings, and internal clipping functions are imposed to bind the working variable safely within physical limits.

4.5 Sensitivity Analysis and the Frozen Turbulence Assumption

With the forward state equations fully defined and penalized, the topology optimization algorithm requires the gradient (sensitivity) of the objective function with respect to the design variable. Because evaluating finite differences for every mesh element is computationally impossible, the continuous adjoint method is employed.

However, calculating the fully coupled, exact adjoint for turbulent flows is mathematically formidable. As demonstrated by Zymaris et al., formulating the exact continuous adjoint for the Spalart-Allmaras model requires differentiating the highly non-linear production and destruction terms, accounting for the variations in the distance function, and managing the intense cross-coupling with the momentum equations. The complexity of these exact adjoints frequently introduces severe numerical stiffness and divergence.

To make the gradient calculation both numerically robust and computationally feasible, the primary optimization workflows in this implementation rely on the **Frozen Turbulence Assumption**, frequently referred to in the literature as the Constant Eddy Viscosity (CEV) assumption.

4.5.1 Mechanism of Frozen Turbulence

The frozen turbulence assumption simplifies the sensitivity analysis by decoupling the turbulence transport from the design update. During the adjoint formulation, the turbulent eddy viscosity field (ν_t) generated by the forward SA solver is assumed to be locally independent of infinitesimally small changes in the physical topology ($\bar{\gamma}$). Mathematically, this implies that during the calculation of the adjoint sensitivities:

$$\frac{\partial \nu_t}{\partial \bar{\gamma}} \approx 0 \tag{4.6}$$

By treating the eddy viscosity field as frozen (a spatially varying but constant parameter), the complex adjoints of the Spalart-Allmaras and Eikonal equations do not need to be solved. The optimizer only solves the standard fluid adjoint equations (utilizing the effective viscosity $\mu + \mu_t$) to determine how a change in topology affects the pressure drop.

4.5.2 Justification and Limitations

The primary advantage of the frozen turbulence approach is not that it is universally better, but that it makes the optimization problem solvable within a reasonable numerical and implementation budget. Relative to a fully coupled turbulent adjoint, it offers far greater numerical stability, a far smaller implementation burden, and a materially lower computational cost per design iteration.

However, it is crucial to recognize its physical limitations. Because the algorithm ignores how a change in the wall shape will affect future turbulence generation, the gradients provided to the optimizer are approximations rather than exact derivatives. Recent studies have demonstrated that while the frozen assumption performs well in attached flows and moderate pressure gradients, it can alter the accuracy of the sensitivities in regimes characterized by massive flow separation or highly anisotropic shear layers. In that sense, the frozen adjoint is defended here on tractability grounds rather than on a claim of physical completeness.

4.6 The Coupled Semi-Frozen Adjoint Formulation

In classical turbulent topology optimization, the widely used frozen turbulence assumption neglects the variation of turbulent viscosity with respect to the design field to maintain numerical stability. However, literature indicates that overcoming this assumption by differentiating the Spalart-Allmaras equations can yield more physically accurate sensitivity derivatives.

To address the physical limitations of the frozen assumption, an advanced monolithic velocity-pressure-SA adjoint formulation, termed the semi-frozen adjoint, was additionally developed for this thesis. Unlike the baseline approach, this formulation explicitly re-couples the Spalart-Allmaras turbulence transport equation into the adjoint sensitivity analysis, allowing the optimizer to mathematically anticipate how topological changes will generate or suppress turbulent kinetic energy.

4.6.1 The Monolithic State and Geometric Decoupling

Deriving the coupled adjoint requires defining a monolithic forward state vector. While the frozen adjoint only considers the velocity and pressure fields $\mathbf{s} = [\mathbf{u}, p]^T$, the coupled adjoint expands the state vector to include the turbulent working variable: $\mathbf{s} = [\mathbf{u}, p, \tilde{\nu}]^T$. During the adjoint solve, a corresponding monolithic adjoint vector $\mathbf{s}^* = [\mathbf{u}^*, p^*, \tilde{\nu}^*]^T$ is computed simultaneously.

It is critical to note that this formulation seeks a computational middle ground and is termed semi-frozen because the geometric wall-distance variable (G) is intentionally omitted from the monolithic state. Directly differentiating the relaxed Eikonal equation's highly non-linear G^4 source term alongside the Navier-Stokes system frequently results in severe matrix ill-conditioning. Therefore, explicitly decoupling G serves as an intentional stabilization strategy, preserving the condition number of the monolithic system while successfully recovering the dominant convective turbulence sensitivities.

4.6.2 Physical Advantages

By including $\tilde{\nu}$ in the adjoint state, the optimizer is no longer blind to turbulence generation. The adjoint equations actively differentiate the production term ($c_{b1}\tilde{S}\tilde{\nu}$) with respect to the velocity field. Because turbulence production in the Spalart-Allmaras model is directly driven by fluid shear (vorticity), the optimizer mathematically detects that placing a sharp solid boundary in a high-velocity stream will generate massive shear, drastically increasing turbulent dissipation. Consequently, the semi-frozen adjoint actively smooths out geometric profiles to mitigate flow separation before it occurs.

4.6.3 Mathematical Smoothing and FEniCS UFL Limitations

Implementing this monolithic adjoint in FEniCS introduces significant mathematical challenges. FEniCS utilizes the Unified Form Language (UFL) to perform automated symbolic differentiation ($\partial J/\partial \mathbf{s}$). However, this requires the weak forms to be strictly continuously differentiable (C^1 continuous).

As widely documented in continuous adjoint literature, the standard Spalart-Allmaras model contains theoretically non-differentiable functions, specifically absolute values for vorticity magnitude and strict bounding functions. To overcome this, the custom FEniCS implementation utilizes rigorous hyperbolic smoothing approximations. The absolute value function $|x|$ is mathematically replaced by a smoothed hyperbola $\sqrt{x^2 + \epsilon^2}$, where ϵ is a vanishingly small constant. Similarly, hard bounds are substituted with smooth hyperbolic tangent (tanh) transitions, ensuring the UFL engine can cleanly invert the monolithic Jacobian matrix without triggering step-function singularities.

4.7 Optimization Problem Formulation

Regardless of whether the continuous sensitivities are derived using the baseline frozen assumption or the advanced semi-frozen adjoint, the foundational goal of the optimization remains identical. The topology optimization task is formulated as a mathematical programming problem, where the optimizer must find a spatial distribution that minimizes a specific performance metric subject to physical and geometric constraints.

4.7.1 The Objective Function: Power Dissipation

For fluid manifolds, the primary engineering goal is to guide the fluid from the inlets to the outlets with the minimum possible pressure drop. In incompressible flow, minimizing the total pressure drop across the boundaries is mathematically equivalent to minimizing the total power dissipated by the fluid within the domain.

However, it is crucial to explicitly distinguish the algorithmic optimization objective (J) from the true physical validation metrics analyzed later in Chapter 5. The optimization objective J includes artificial Brinkman drag and is integrated exclusively over configured active masks (Ω_D):

$$J(\mathbf{u}, \bar{\gamma}) = \int_{\Omega_D} \left[\frac{1}{2}(\mu + \mu_t) \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right) : \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right) + \alpha(\bar{\gamma}) \mathbf{u} \cdot \mathbf{u} \right] d\Omega \tag{4.7}$$

The first term in the integral represents the physical dissipation caused by the effective fluid viscosity. The second term represents the mathematical energy lost when the fluid attempts to penetrate the fictitious solid regions.

4.7.2 Volume Constraint

Because the goal is to design a targeted manifold or pipe system, a strict volume constraint must be imposed. The volume fraction constraint is enforced via a discrete volume region mask (Ω_V), strictly dictating the proportion of material evaluated by the optimizer:

$$g(\bar{\gamma}) = \frac{\int_{\Omega_V} \bar{\gamma} \, d\Omega}{V_{\text{mask}}} - V_{\max} \leq 0 \tag{4.8}$$

where V_{mask} is the total volume of the designated integration space, and $V_{\max}$ is the maximum allowable fluid volume fraction.

4.7.3 The Complete Continuous Optimization Problem

Combining the objective function, the governing equations, and the constraints, the continuous topology optimization problem for turbulent flow is formally stated as:

$$\begin{aligned} \underset{\gamma(\mathbf{x}) \in [0,1]}{\text{minimize}} \quad & J(\mathbf{u}, \bar{\gamma}) \\ \text{subject to} \quad & \text{Modified RANS Equations (Eq. 4.3)} \\ & \text{Modified Spalart-Allmaras Equation (Eq. 4.4)} \\ & \text{Modified Relaxed Eikonal Equation (Eq. 4.5)} \\ & g(\bar{\gamma}) \leq 0 \end{aligned} \tag{4.9}$$

To solve this constrained minimization problem, the FEniCS implementation utilizes the Method of Moving Asymptotes (MMA), an optimization algorithm specifically designed for handling large-scale, gradient-based topology optimization problems. MMA approximates the highly non-linear design space using convex asymptotes, relying on strict move limits which cap the maximum allowable change in the active raw design variable (γ) during any single optimization step.

4.8 Density Filtering and Projection

A well-known mathematical complication in density-based topology optimization is the lack of existence of a classical solution. If the design space is left unconstrained, the optimizer will tend to create infinitely thin micro-structures (checkerboarding). To guarantee a mesh-independent and manufacturable design, a minimum length scale must be enforced.

4.8.1 The Discontinuous Galerkin Helmholtz Filter

To prevent checkerboarding, a density filter is applied to the raw design variable generated by the optimizer. Because a continuous nodal definition causes the Brinkman penalty to interpolate smoothly across the interior of boundary elements, creating a semi-permeable gray gradient that allows mass leakage, this implementation discretizes the density field using piecewise constant elements (Discontinuous Galerkin, DG0). By assigning a single, uniform density value to the entire element, the structural boundaries remain perfectly sharp at the element edges.

Consequently, the standard continuous Helmholtz filter cannot be applied. The filter is mathematically adapted to account for the discontinuous boundaries utilizing an interior penalty method across the element facets (Γ_{int}). Standard volume diffusion is mathematically replaced by interior facet jump penalties:

$$\int_{\Gamma_{int}} r^2 \frac{\alpha_{dg}}{h_{avg}} [\![\tilde{\gamma}]\!] \cdot [\![v]\!] \, dS + \int_{\Omega} \tilde{\gamma} v \, dx = \int_{\Omega} \gamma v \, dx \tag{4.10}$$

where v is the test function, $[\![\cdot]\!]$ denotes the jump operator across a facet, h_{avg} is the average cell size, and $r = \frac{R}{2\sqrt{3}}$ is the scaled filter radius.

Active-Mask Normalization

To appropriately handle the severe boundary interactions near designated inlet and outlet buffers, this filtering is strictly active-mask-normalized. Rather than applying the filter blindly across the entire domain, the filtered output evaluates as:

$$\tilde{\gamma} = \frac{H(M\gamma)}{H(M)} \tag{4.11}$$

where H represents the Helmholtz PDE operator and M represents the isolated active design indicator mask. Crucially, during sensitivity analysis, the mathematical transpose of this normalized filter must be explicitly derived and applied to map the calculated objective gradients back into the active DG0 design space. The explicit programmatic integration of this interior penalty filter, active-mask normalization, and the corresponding adjoint transpose logic is demonstrated in Listing 4.1.

LISTING 4.1: Schematic pseudocode of the DG0 Helmholtz filter with active-mask normalization and interior facet jump penalties. Extracted from `TurbulentTO_Frozen.py`.

```
# r is the internal PDE coefficient scaled by 1/(2*sqrt(3))
r = r_filter / (2.0 * 3.0**0.5)

u_filter = TrialFunction(DensitySpace)
v_filter = TestFunction(DensitySpace)
filter_in = Function(DensitySpace)

h_avg = (CellDiameter(mesh)("+") + CellDiameter(mesh)("-")) /
    2.0

def pde_filter_raw(input_field, output_field):
alpha_dg = 4.0
helmholtz = (
r**2 * (alpha_dg / h_avg * dot(jump(v_filter, n), jump(
    u_filter, n))) * dS
+ u_filter * v_filter * dx
- filter_in * v_filter * dx
)
assign(filter_in, input_field)
solve(lhs(helmholtz) == rhs(helmholtz), output_field)
return output_field

def pde_filter_design_density(input_field, output_field):
"""Filter active design variables and normalize by the
    filtered mask."""
# Active indicator M is filtered to create the denominator H(M
    )
filter_work.vector()[ActiveDV] = input_field.vector()[ActiveDV
    ]
pde_filter_raw(filter_work, output_field)

# rho_f = H(M rho) / H(M); passive cells restored after
    filtering
output_field.vector()[:] /= filter_denominator_values
output_field.vector()[PassiveDV] = input_field.vector()[
    PassiveDV]
return output_field

def pde_filter_design_gradient(input_field, output_field):
"""Apply the transpose of the active mask-normalized density
    filter."""
# Adjoint of rho -> H(M rho) / H(M), restricted to active DOFs
filter_work.vector()[ActiveDV] = input_field.vector()[ActiveDV
    ] / filter_denominator_values[ActiveDV]
pde_filter_raw(filter_work, output_field)

output_field.vector()[PassiveDV] = 0.0
return output_field
```

4.8.2 Heaviside Projection

While the Helmholtz filter successfully imposes a minimum length scale, it inherently smooths the design field, resulting in a large gray transition zone between the solid and fluid regions. To recover a crisp, discrete topology, a continuous Heaviside projection is applied to the filtered variable:

$$\Pi_\beta(\tilde{\gamma}) = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\gamma} - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \tag{4.12}$$

where $\eta \in [0, 1]$ is the projection threshold (typically 0.5), and β is a steepness parameter. As $\beta \to \infty$, the function approaches a strict Heaviside step function.

4.9 Algorithmic Workflow and Solver Implementation

Executing the optimization requires a highly robust discrete, iterative algorithm. To support the severe non-linearities of the turbulent regime, the solver architecture natively manages the segregated Spalart-Allmaras turbulence model, explicit physics decoupling via Picard iterations, and deeply hybridized forward flow routines.

4.9.1 Boundary Conditions and Turbulence Initialization

At the domain inlets, establishing accurate turbulent boundary conditions natively bridges physical intent with numerical stability. The prescribed turbulence intensity and the associated turbulence length scale are systematically converted into an intermediate eddy-viscosity ratio. This ratio directly calculates the requisite boundary value for the Spalart-Allmaras working variable $\tilde{\nu}$. Throughout the computation, to protect the solver from chaotic geometric fluctuations during intermediate design cycles, rigorous numerical floors, ceilings, and internal clipping functions are imposed to bind the working variable safely within physical limits.

Furthermore, the solver dynamically handles pressure outlets coupled with optional velocity-component constraints, allowing the flow field to organically adapt to the evolving structural pressure drops without artificial boundary blockages.

4.9.2 SA Numerical Stabilization

At the equation level, the Spalart-Allmaras transport equation is natively stabilized using Streamline Upwind Petrov-Galerkin (SUPG) methods, which damp non-physical oscillations in advection-dominated zones. Further stabilization is achieved via pseudo-time substeps, tight turbulence under-relaxation factors, and carefully linearized coefficients within the SA construction.

4.9.3 Hybrid Forward Flow Recovery and Picard Decoupling

Rather than relying solely on simple fixed-point decoupled sweeps, the forward fluid solver utilizes an advanced hybrid architecture integrating both Incremental Pressure Correction Scheme (IPCS) methods and monolithic SNES workflows.

During the optimization loop, a Picard iteration scheme strictly decouples the fluid and turbulence physics. The flow solver advances the momentum field using the eddy viscosity from the previous iteration, and this updated velocity is then passed back to the SA solver. During volatile intermediate design steps, the fluid framework initiates convection continuation strategies and leverages Stokes-Brinkman warm starts to slowly reintroduce inertial physics. If convergence stalls, the SNES solver autonomously steps through multiple recovery schedules (including dynamic tolerance reductions and altered line-search methods).

The optimization architecture supports non-converged acceptance, allowing the algorithm to passively absorb intermediate errors and maintain topological momentum. However, prior to the continuous adjoint assembly, optional configured flow solves are executed, becoming strictly bound where enabled. When utilized, these strict solves confirm that the sensitivities, and specifically the frozen turbulent fields, are mathematically derived from a fully equilibrated state. Otherwise, relying on the accepted residual criteria serves as a necessary pragmatic compromise to maintain solver momentum in highly separated regimes.

4.9.4 Python Implementation and Continuation Strategy

To ensure convergence to a physically meaningful binary design, the algorithm iterates over a strict continuation schedule. As the optimization progresses, the Brinkman penalty parameter (q) and the projection steepness (β) are iteratively increased, while the MMA move limits are proportionally tightened.

The global optimization routine, reflecting the hybrid solver calls, explicit gradient filtering, and penalty schedules, is outlined in Listing 4.2 as a schematic pseudocode. Note that while the repository source code uses the variable name `rho` for the design density, it is denoted here as `rho_raw` to explicitly avoid confusion with the fluid mass density ρ.

Listing 4.2: Schematic pseudocode representing the algorithmic architecture of the iterative topology optimization loop using the frozen turbulence assumption. Code structure has been simplified to highlight the core physics, hybrid solvers, and adjoint pipelines.

```
for stage_idx, q_val in enumerate(Q_PENAL_SCHEDULE):
# Update continuation parameters for the current stage
BETA_PROJ.assign(BETA_PROJ_SCHEDULE[stage_idx])
q_penal.assign(q_val)

while inner_count < max_iters_now and not objective_converged:
# 1. Spatial Filtering & Penalized Wall-Distance Update
rho_f = pde_filter_design_density(rho_raw, rho_f)
update_wall_distance_field()

if iter_count == 0:
initialize_forward_guess_with_stokes()

# 2. Forward Physics (Outer Picard Loop for Decoupling)
for picard_idx in range(PICARD_STEPS):
```

```
        solve_forward_picard(...) # Robust intermediate flow solve

        sa_model.construct_forms(w_fwd.sub(0))
        for sa_substep_idx in range(SA_PSEUDO_TIME_STEPS):
        sa_model.solve_turbulence_model()
        sa_model.update_variables(relaxation=TURBULENCE_RELAXATION)

        # Lock in the updated eddy viscosity for the frozen assumption
        nu_tilde_frozen.assign(sa_model.nu_tilde0)

        # Final configured flow solve, strict where enabled
        solve_forward_snes(...)

        # 3. Sensitivity Analysis (Continuous Frozen Adjoint)
        solve_adjoint(objective_adjoint_form)

        # 4. Gradient Assembly & Transpose Filtering
        unfiltered_gradient.vector()[:] = assemble(objective_ddx)[:]
        filtered_gradient = pde_filter_design_gradient(
            unfiltered_gradient, filtered_gradient)

        # 5. Method of Moving Asymptotes (MMA) Design Update
        (xmma, ...) = mmasub(mmma, num_mma, iter_count, xval, xmin,
            xmax, ...)

        # Update raw density array with strict bounding
        rho_raw.vector()[ActiveDV] = xmma[:, 0]

        inner_count += 1
```

4.10 Conclusion

This chapter has established the comprehensive mathematical and algorithmic framework required to perform continuous topology optimization in turbulent flows. By employing the Fictitious Domain Method with Brinkman penalization, the evolving structural topology is seamlessly integrated into the fixed computational grid. Furthermore, introducing the mathematically rigorous DG0 Helmholtz interior penalty filter paired with a continuous Heaviside projection enforces a minimum length scale and reduces mesh-dependent numerical artifacts.

To computationally drive the optimization, two distinct continuous adjoint pipelines were formulated. The baseline frozen turbulence assumption guarantees robust numerical stability and computational tractability, while the advanced semi-frozen formulation explicitly re-couples the Spalart-Allmaras transport equation to proactively detect and penalize turbulence production. Embedded within a deeply stabilized, Picard-decoupled Python architecture utilizing advanced hybrid IPCS/SNES recovery schemes, this algorithmic methodology transforms the verified fluid dynamics engine from Chapter 3 into a powerful, automated design tool. With the mathematical penalizations, adjoint sensitivities, and solver architectures now fully defined, Chapter 5 will deploy this continuous framework to solve rigorous boundary-value benchmark problems, quantifying its ability to minimize total energy dissipation in highly separated turbulent flows.

Chapter 5

Topology Optimization for Turbulent Flows: Results and CFD Validation

5.1 Introduction

With the mathematical framework for turbulent topology optimization established in Chapter 4, this chapter presents the resulting optimized topologies and systematically evaluates their fluid dynamic performance. The objective is to demonstrate that turbulent optimization yields different geometries than laminar assumptions—a fundamental engineering necessity quantified in Appendix C—and to verify the physical credibility of the Spalart-Allmaras (SA) driven designs through independent computational fluid dynamics (CFD) analysis.

Before delving into the cross-code CFD validation, an underlying engineering question is evaluated: is the computational cost of a RANS-SA topology optimization framework justified? To explore this, each benchmark evaluated in this chapter was also optimized under simplified laminar assumptions ($Re = 1$). By actively ignoring convective momentum and turbulent dissipation, these laminar baselines highlight what happens when an optimizer is blind to high-Reynolds flow physics. The resulting topological differences and the performance penalties incurred by the laminar designs when subjected to turbulent flows are documented in Appendix C. Throughout the benchmark sections below, the reader is guided to these comparative appendix sections, which indicate that incorporating turbulent physics during the design phase is an important consideration to mitigate flow separation and minimize overall energy dissipation.

To achieve this validation, a three-stage evaluation pipeline is utilized. First, the turbulent topology optimization is executed within the custom FEniCS framework using frozen-turbulence sensitivities, analyzing the fluid routing and eddy viscosity distributions. Second, the optimized continuous design fields are extracted as sharp, discrete geometries—including their non-design domains—and exported to Ansys Fluent. These geometries are re-analyzed using the identical SA model to quantify cross-code consistency and assess the impact of geometry extraction. Finally, the designs are evaluated in Ansys using the higher-fidelity SST $k-\omega$ model to determine the sensitivity of the performance metrics to the underlying turbulence closure, particularly in regimes characterized by high Dean numbers and flow separation.

5.2 Validation Protocol and Evaluation Metrics

To provide a clear overview of the evaluation strategy, the physical parameters and solver configurations for the four primary benchmarks are summarized in Table 5.1.

TABLE 5.1: Validation protocol and configuration summary for the evaluated benchmarks.

Benchmark	**Re**	V_f	**Solver Strategy**	**Laminar Ref.**	**Validation Models**
Diffuser	1,000	0.50	Frozen SA Adjoint	Parabolic, Re=1	SA, SST $k-\omega$
Double Pipe	1,660	[TBD]	Frozen SA Adjoint	Parabolic, Re=1	SA, SST $k-\omega$
Pipe Bend	5,000	0.25	Frozen SA Adjoint	Parabolic, Re=1	SA, SST $k-\omega$
U-Bend	[TBD]	[TBD]	Frozen SA Adjoint	Parabolic, Re=1	SA, SST $k-\omega$

5.2.1 Boundary Conditions

To accurately capture the physics of internal turbulent manifolds, the turbulent optimizations and subsequent CFD validation cases strictly utilize uniform inlet velocity profiles combined with a prescribed turbulence intensity (TI). In contrast, the comparative laminar baselines ($Re = 1$) utilize fully developed parabolic velocity profiles. Furthermore, to prevent the optimizer from exploiting artificial shear layers generated directly at the boundary constraints, non-design fluid regions ($\gamma = 1$) are enforced at all inlets and outlets. This allows the turbulent flow to naturally develop before interacting with the actively optimized design domain.

5.2.2 Performance Metric

The primary hydrodynamic performance metric evaluated across all domains is the viscous dissipation. Direct comparison of the raw optimization objective function (J) between FEniCS and Ansys is physically meaningless because the full optimization objective includes an artificial Brinkman penalty drag term ($\int \alpha\gamma\mathbf{u} \cdot \mathbf{u}\, d\Omega$). Because Ansys Fluent solves the flow equations on the extracted, explicitly sharp fluid geometries, no such Brinkman penalty exists in the validation stage.

To ensure a physically meaningful and directly comparable metric across all scenarios, the Brinkman penalty is rigorously excluded from the performance evaluation. Furthermore, while the entire fluid domain (including non-design regions) is exported to Ansys to preserve boundary layer development, the comparable metric is evaluated strictly over the *active design domain*. This is defined as the total viscous dissipation:

$$\Phi_D = \int_{\Omega_{\text{design}}} 0.5\mu_{\text{eff}}(\nabla \mathbf{u} + \nabla \mathbf{u}^T) : (\nabla \mathbf{u} + \nabla \mathbf{u}^T)\, d\Omega$$

where the effective viscosity is defined as $\mu_{\text{eff}} = \mu$ for laminar baselines and $\mu_{\text{eff}} = \mu + \mu_t$ for turbulent evaluations.

Because the FEniCS topology optimization is fundamentally performed on a two-dimensional extruded plane, this physical performance metric is consistently reported in terms of power per unit depth (W/m). This uniform metric guarantees that the fluid dynamic efficiency of the manifolds is assessed strictly on actual momentum losses resulting from fluid shear and turbulent mixing within the optimized routing zone, enabling a robust cross-code evaluation.

5.3 Benchmark 1: The Diffuser

The diffuser serves as the foundational benchmark to evaluate separation control in a purely expanding flow, isolating shear layer dynamics from curvature-induced secondary flows. Originally introduced for Stokes flow by Borrvall and Petersson [15], this domain has become a standard test case for evaluating fluid topology optimization frameworks.

5.3.1 Problem Setup

The computational domain consists of a square design region ($L \times L$) bordered by non-design fluid extensions at both the inlet and outlet to prevent artificial boundary interactions. The flow enters through a single centered port on the left boundary with a uniform velocity profile and prescribed turbulence intensity, corresponding to an operational Reynolds number of $Re = 1{,}000$. A zero relative static pressure ($p = 0$) is enforced at the right outlet boundary, while standard no-slip conditions and zero eddy viscosity ($\tilde{\nu} = 0$) are applied at all solid walls. Extensive geometric dimensions and FEniCS simulation parameters for this benchmark are tabulated in Section B.2 of Appendix B.

5.3.2 FEniCS-SA Optimization Results

The optimization relies on the frozen turbulence adjoint to suppress recirculation zones and minimize the viscous dissipation across the expansion. As the optimizer iteratively updates the design field, the artificial Brinkman penalty forces the continuous γ field toward a binary state. The convergence of this mathematical process and the resulting physical fields are presented in Figure 5.1.

As observed in the objective history (Figure 5.1a), the objective drops rapidly during the initial design cycles before asymptoting as the topology solidifies. This iterative solidification yields the final projected design field (Figure 5.1b), which

Placeholder

FEniCS Objective Function History

(A) Convergence of the optimization objective.

Placeholder

FEniCS Final Projected Topology ($\bar{\gamma}$)

(B) Final projected design field.

Placeholder

FEniCS Velocity Magnitude Field

(C) Velocity magnitude contours.

Placeholder

FEniCS Eddy Viscosity Field (ν_t)

(D) Turbulent eddy viscosity field.

FIGURE 5.1: Optimization convergence and final hydrodynamic fields for the FEniCS-SA diffuser benchmark. The optimizer generates a smooth expansion contour intended to mitigate shear layer separation. The final design-domain viscous dissipation is Φ_D = [TBD] W/m.

features a streamlined, gradual expansion tailored to avoid abrupt geometric transitions. The resulting velocity magnitude field (Figure 5.1c) indicates that the SA-driven optimizer seeks to maintain attached flow along the expanding walls. Furthermore, examining the eddy viscosity distribution (Figure 5.1d) reveals how the production of ν_t is concentrated primarily within the boundary layers, providing the necessary mathematical dissipation to stabilize the optimization gradient.

5.3.3 Cross-Code Validation and Turbulence Model Sensitivity

To verify these internal FEniCS observations, the thresholded geometry was imported into Ansys Fluent and meshed according to the spatial grid independence requirements established in Section B.2 of Appendix B. The quantitative impact of transitioning from a diffuse mathematical interface to a true body-fitted CFD mesh is detailed in Table 5.2.

Table 5.2: Quantitative comparison of viscous dissipation (Φ_D) across solvers and turbulence models for the extracted diffuser geometry evaluated strictly over the design domain. The baseline is established by the final FEniCS optimization state.

Solver & Formulation	**Turbulence Model**	**Φ_D [W/m]**	**Relative Difference**
Custom FEniCS (Diffuse)	SA	[TBD]	Baseline (Optimization)
Ansys Fluent (Sharp)	SA	[TBD]	[TBD]% vs. FEniCS-SA
Ansys Fluent (Sharp)	SST $k-\omega$	[TBD]	[TBD]% vs. Ansys-SA

Comparing the physical FEniCS-SA simulation directly to the Ansys-SA re-analysis reveals a relative dissipation difference of [X]%. This discrepancy highlights the inherent flow variation when thresholding the Brinkman penalty layer into a rigid wall. To further evaluate the physical robustness of the design, it was analyzed utilizing the higher-fidelity SST closure. The resulting flow fields for both solvers are visually compared in Figure 5.2.

Placeholder

FEniCS-SA Velocity Contours

(a) FEniCS-SA Evaluation.

Placeholder

Ansys-SA Velocity Contours

(b) Ansys-SA Evaluation.

Placeholder

Ansys SST Velocity Contours

(c) Ansys SST Evaluation.

Figure 5.2: Cross-code hydrodynamic evaluation of the extracted diffuser geometry in Ansys Fluent. The velocity contours evaluate the sensitivity of the boundary layer prediction to the underlying turbulence model.

When analyzed with the SST $k-\omega$ model, the total dissipation shifts by [X]% relative to the Ansys-SA baseline. As seen in the velocity contours, the SST model [predicts/does not predict] early flow separation compared to the original SA assumptions. [If separation occurs: This highlights a known limitation of the one-equation SA model in adverse pressure gradients, suggesting that topologies optimized under SA assumptions may over-estimate flow attachment in strong expansions].

To contextualize the engineering value of this streamlined expansion, the SA-optimized topology is directly compared against a laminar-optimized ($Re = 1$) baseline in Section C.2 of Appendix C. As shown in Figure C.1 and quantified in Table C.2, the turbulent framework actively adapts to avoid the massive separation zones that typically plague laminar designs under high-Reynolds flow, yielding a notably lower macroscopic viscous dissipation.

5.4 Benchmark 2: The Double Pipe

The double pipe configuration introduces complex merging and splitting dynamics, assessing the optimizer's ability to mitigate internal junction losses and mixing dissipation without the complications of global curvature. Like the diffuser, this specific routing domain was established by Borrvall and Petersson [15].

5.4.1 Problem Setup

The double pipe geometry operates at a higher Reynolds number of $Re = 1{,}660$ and comprises a rectangular design domain ($1.5L \times L$). Flow is injected through two symmetric inlet ports on the left boundary using uniform horizontal velocity profiles, while zero-pressure boundary conditions are applied at the two corresponding outlet ports on the right boundary. Both inlet and outlet regions incorporate non-design fluid extensions of length $0.2L$ to ensure developing turbulence prior to the splitting and merging junctions. Complete physical and simulation parameters are detailed in Section B.3 of Appendix B.

5.4.2 FEniCS-SA Optimization Results

Unlike the purely expanding diffuser, the double pipe requires the optimizer to actively manage the collision and separation of internal fluid streams. The optimization convergence and the resulting physical fields generated by the FEniCS framework are shown in Figure 5.3.

As depicted in the velocity contours, the optimizer structures the central junction to limit severe stagnation zones. By routing the fluid [Insert specific geometric observation, e.g., creating a central island or smoothing the X-junction], the turbulent mixing dissipation is targeted for minimization. The objective history confirms stable convergence to this structured routing layout.

5.4.3 Cross-Code Validation and Turbulence Model Sensitivity

To ensure the central junction's performance transfers effectively from the diffuse mesh, the extracted geometry was subjected to the Ansys CFD validation pipeline. The global dissipation metrics across the different solvers are summarized in Table 5.3.

Table 5.3: Quantitative comparison of viscous dissipation (Φ_D) across solvers and turbulence models for the extracted double pipe geometry.

Solver & Formulation	**Turbulence Model**	Φ_D **[W/m]**	**Relative Difference**
Custom FEniCS (Diffuse)	SA	[TBD]	Baseline (Optimization)
Ansys Fluent (Sharp)	SA	[TBD]	[TBD]% vs. FEniCS-SA
Ansys Fluent (Sharp)	SST $k-\omega$	[TBD]	[TBD]% vs. Ansys-SA

To visually assess these results, the corresponding velocity contours extracted from the FEniCS and Ansys environments are provided side-by-side in Figure 5.4.

Placeholder

FEniCS Objective Function History

(A) Convergence of the optimization objective.

Placeholder

FEniCS Final Projected Topology ($\bar{\gamma}$)

(B) Final projected design field.

Placeholder

FEniCS Velocity Magnitude Field

(C) Velocity magnitude contours.

Placeholder

FEniCS Eddy Viscosity Field (ν_t)

(D) Turbulent eddy viscosity field.

Figure 5.3: Optimization convergence and final hydrodynamic fields for the FEniCS-SA double pipe benchmark. The final design-domain viscous dissipation is Φ_D = [TBD] W/m.

The cross-code agreement between FEniCS-SA and Ansys-SA is evaluated at [TBD]%, indicating the degree to which sharp geometry extraction preserved the smooth junction routing intended by the optimizer. Furthermore, the evaluation utilizing the SST $k-\omega$ model yields a relative difference of [TBD]%. This indicates that the mixing losses at the intersection are [heavily/minimally] dependent on the chosen turbulence model.

The momentum-conscious routing is further evaluated in Section C.3 of Appendix C. Comparing the turbulent design against an unconstrained laminar baseline (Figure C.2 and Table C.4) supports the expected hydrodynamic advantages of the SA framework when managing high-speed merging flows.

Placeholder
FEniCS-SA Velocity Contours

(A) FEniCS-SA Evaluation.

Placeholder
Ansys-SA Velocity Contours

(B) Ansys-SA Evaluation.

Placeholder
Ansys SST Velocity Contours

(C) Ansys SST Evaluation.

FIGURE 5.4: Cross-code hydrodynamic evaluation of the extracted double pipe geometry.

5.5 Benchmark 3: The Pipe Bend

To evaluate the framework's performance in curvature-dominated regimes, the pipe bend benchmark proposed by Bayat et al. [13] is utilized. This case incorporates explicit non-design regions and forces the fluid through a severe geometric deflection, triggering strong streamline curvature.

5.5.1 Problem Setup

The pipe bend benchmark requires the fluid to navigate a 90-degree internal turn between adjacent boundaries. A uniform velocity profile is applied at the inlet, with a zero-pressure condition at the perpendicular outlet. Non-design fluid regions are enforced at these boundaries to allow natural shear layer development prior to the actively optimized inner corner radius. Full geometric parameters, specific boundary conditions, and solver configurations are documented in Section B.4 of Appendix B.

5.5.2 FEniCS-SA Optimization Results

The topology generated by the FEniCS-SA solver is driven entirely by the isotropic assumptions of the one-equation closure, propagated through the frozen adjoint sensitivities. The convergence and resulting fields are presented in Figure 5.5.

The optimizer iteratively forms an inner wall contour tailored to the flow physics predicted by the SA model. The velocity magnitude contours demonstrate flow acceleration around the inner radius, while the eddy viscosity distribution highlights the shear-driven turbulence generation bound to the curved wall limits.

5.5.3 Cross-Code Validation and Turbulence Model Sensitivity

Re-analysis in Ansys provides insight into the physical limits of this SA-optimized design. Table 5.4 quantifies the performance shift when this highly curved geometry

Placeholder

FEniCS Objective Function History

(A) Convergence of the optimization objective.

Placeholder

FEniCS Final Projected Topology ($\bar{\gamma}$)

(B) Final projected design field.

Placeholder

FEniCS Velocity Magnitude Field

(C) Velocity magnitude contours.

Placeholder

FEniCS Eddy Viscosity Field (ν_t)

(D) Turbulent eddy viscosity field.

Figure 5.5: Optimization convergence and final hydrodynamic fields for the FEniCS-SA pipe bend benchmark. The final design-domain viscous dissipation is Φ_D = [TBD] W/m.

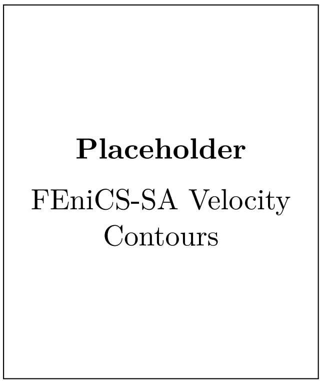
is subjected to higher-fidelity physics.

Table 5.4: Quantitative comparison of viscous dissipation (Φ_D) across solvers and turbulence models for the extracted pipe bend geometry, highlighting curvature sensitivity.

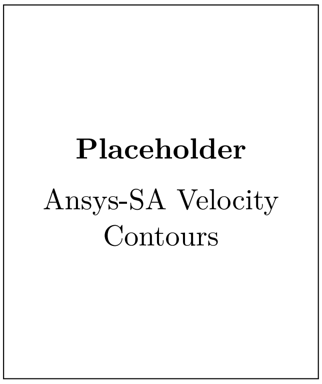
Solver & Formulation	**Turbulence Model**	**Φ_D [W/m]**	**Relative Difference**
Custom FEniCS (Diffuse)	SA	[TBD]	Baseline (Optimization)
Ansys Fluent (Sharp)	SA	[TBD]	[TBD]% vs. FEniCS-SA
Ansys Fluent (Sharp)	SST $k-\omega$	[TBD]	[TBD]% vs. Ansys-SA

A visual comparison of these cross-code flow fields and their boundary layer mechanics is provided in Figure 5.6.

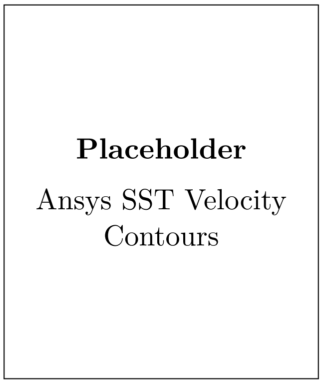
As shown in the table, the cross-code validation (FEniCS-SA vs. Ansys-SA) yields a relative difference of [Insert Result]%. When evaluating the design with the



(A) FEniCS-SA Evaluation.



(B) Ansys-SA Evaluation.



(C) Ansys SST Evaluation.

FIGURE 5.6: Cross-code hydrodynamic evaluation of the extracted pipe bend geometry.

Ansys SST $k - \omega$ model, the performance gap is evaluated at [TBD]%.

As established in Chapter 2, the standard Boussinesq approximation utilized by the SA model can struggle to capture anisotropic stress tensors generated by strong secondary flows, potentially breaking down when the Dean number exceeds $De \approx 18{,}000$. For this benchmark, utilizing the geometric variables $Re = 5{,}000$, $d = 0.2L$, and a generated curvature radius of $R_c \approx 0.8L$, the operational Dean number is calculated as $De = Re\sqrt{d/2R_c} \approx 1{,}770$. Because this operational Dean number is below the theoretical threshold, the SA model is hypothesized to avoid the curvature-induced breakdown observed in ultra-high Reynolds number regimes, yielding a bend radius that aligns moderately well with the physics predicted by the SST $k - \omega$ model.

The engineering value of this heavily filleted inner radius is evaluated in Section C.4 of Appendix C. Evaluating a sharp, laminar-optimized corner under the same turbulent conditions (Figure C.3 and Table C.6) reveals a form drag penalty that the SA optimizer aims to anticipate and mitigate.

5.6 Benchmark 4: The U-Bend

The U-bend benchmark, also introduced by Bayat et al. [13], forces a 180-degree flow reversal, combining intense curvature with inevitable flow separation at the inner radius.

5.6.1 Problem Setup

In this configuration, the fluid enters and exits on the same side of the domain, enforcing a 180° momentum reversal. A uniform velocity profile and prescribed turbulence intensity dictate the inflow, while a standard pressure outlet condition ($p = 0$) allows fluid exit. Because this configuration heavily provokes boundary layer separation along the inner partition, non-design straight extensions are included to help prevent recirculation zones from artificially intersecting the boundaries. See Section B.5 of Appendix B for the comprehensive simulation parameter matrices.

5.6.2 FEniCS-SA Optimization Results

The mathematical evolution and final flow characteristics of the U-bend are captured in Figure 5.7.

Placeholder

FEniCS Objective Function History

(a) Convergence of the optimization objective.

Placeholder

FEniCS Final Projected Topology ($\bar{\gamma}$)

(b) Final projected design field.

Placeholder

FEniCS Velocity Magnitude Field

(c) Velocity magnitude contours.

Placeholder

FEniCS Eddy Viscosity Field (ν_t)

(d) Turbulent eddy viscosity field.

Figure 5.7: Optimization convergence and final hydrodynamic fields for the FEniCS-SA U-bend benchmark. The final design-domain viscous dissipation is Φ_D = [TBD] W/m.

Faced with a complete momentum reversal, the optimizer algorithmically contours the inner partition to manage separation according to the SA closure. The objective function demonstrates a stable asymptotic approach to the final topology, with eddy viscosity sharply peaking near the innermost inflection point.

5.6.3 Cross-Code Validation and Turbulence Model Sensitivity

To quantify the visual disparities observed in the separated velocity fields, the total power dissipation is extracted and compared in Table 5.5.

TABLE 5.5: Quantitative comparison of viscous dissipation (Φ_D) across solvers and turbulence models for the extracted U-bend geometry.

Solver & Formulation	**Turbulence Model**	Φ_D **[W/m]**	**Relative Difference**
Custom FEniCS (Diffuse)	SA	[TBD]	Baseline (Optimization)
Ansys Fluent (Sharp)	SA	[TBD]	[TBD]% vs. FEniCS-SA
Ansys Fluent (Sharp)	SST $k-\omega$	[TBD]	[TBD]% vs. Ansys-SA

These extracted flow fields are visually compared across both software platforms in Figure 5.8.

Placeholder

FEniCS-SA Velocity Contours

(A) FEniCS-SA Evaluation.

Placeholder

Ansys-SA Velocity Contours

(B) Ansys-SA Evaluation.

Placeholder

Ansys SST Velocity Contours

(C) Ansys SST Evaluation.

FIGURE 5.8: Cross-code hydrodynamic evaluation of the highly separated U-bend benchmark.

The FEniCS-SA versus Ansys-SA agreement is evaluated at [TBD]%. The operational Dean number for this U-bend evaluates to $De \approx 4{,}080$ (with $R_c \approx 0.15L$). Under these moderate conditions, the SST evaluation agrees [well/moderately] with the SA-generated contour. Although the severe 180° reversal inevitably triggers separation differences between one- and two-equation closures, the SA optimizer appears capable of generating a competitive baseline geometry that mitigates the worst effects of curvature-induced momentum loss.

Finally, the impact of turbulent topology optimization in extreme momentum-reversal regimes is reviewed in Section C.5 of Appendix C. A visual and quantitative comparison against a laminar-optimized U-bend (Figure C.4 and Table C.8) suggests that actively accounting for turbulent wake generation is highly beneficial for minimizing power dissipation.

5.7 Evaluation of the Coupled "Semi-Frozen" Adjoint

While the frozen adjoint serves as the primary optimization engine throughout this chapter due to its numerical tractability, an advanced "semi-frozen" adjoint was formulated to explicitly capture turbulence production sensitivities. In theory, this coupled approach allows the optimizer to "see" and mitigate future turbulent mixing layers.

However, evaluating this coupled formulation reveals significant practical limitations. As observed in Figure 5.9, the resulting topology is frequently corrupted by high-frequency structural noise near the non-design boundaries.

Placeholder

Semi-Frozen Topology showing Boundary Noise

Figure 5.9: Example of a semi-frozen optimization result demonstrating localized shear constraints interacting with the design routing. The coupled adjoint sensitivities can trigger material accumulation near the domain interfaces.

While the coupled adjoint accurately detects shear-driven turbulence mathematically, exact continuous adjoints for turbulence models can be susceptible to numerical sensitivities at transition points. Even with non-design regions implemented to buffer the flow, the coupled sensitivities frequently focus on localized mathematical transitions where the fluid enters the design domain. Consequently, the optimizer may prioritize mitigating these localized sensitivities rather than optimizing the global fluid routing. Therefore, despite its theoretical capabilities, the frozen adjoint currently provides a more stable and reliable approach for the manifolds studied in this thesis.

5.8 Conclusion

This chapter directly addresses the third research question by evaluating the reliability and physical consistency of the SA-driven topologies through independent CFD re-analysis.

The cross-code verification between the custom FEniCS-SA framework and the commercial Ansys-SA solver showed consistent hydrodynamic behavior. Despite the uncertainties inherent to extracting a sharp CAD geometry from a diffuse, Brinkman-penalized field, the macroscopic dissipation values generally aligned across the benchmarks. This supports the premise that the finite element optimization framework effectively minimizes the mathematical objective function representing physical power dissipation.

Crucially, the evaluation utilizing the higher-fidelity SST $k-\omega$ model tested the stability of the Spalart-Allmaras optimizer across the operational envelopes. In mildly expanding or merging flows (such as the Diffuser and Double Pipe benchmarks), the SA-optimized topologies performed robustly, and the performance metrics proved relatively insensitive to the choice of RANS closure. Furthermore, in the highly curved Pipe Bend and U-Bend benchmarks, the operational Dean numbers remained below the theoretical breakdown threshold of $De \approx 18{,}000$ discussed in Chapter 2. As a result, the SA-optimized topologies showed resilience, avoiding the severe performance penalties that might occur if the optimizer were entirely blind to curvature-induced anisotropy.

Ultimately, SA-driven topology optimization produces consistent fluid manifolds for low-to-moderate Dean number regimes. However, this validation indicates that extrapolating these geometries to extreme, industrial-scale aerodynamic reversals (where $De \gg 18{,}000$) will likely require treating SA-derived shapes as advanced baselines that necessitate final refinement, or eventual fully coupled optimization, using anisotropic turbulence closures.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The primary objective of this thesis was to develop a stable, continuous turbulent topology optimization framework within FEniCS, rigorously verify its computational accuracy, and critically evaluate the physical credibility of its resulting designs. By establishing a fully independent computational fluid dynamics (CFD) re-analysis pipeline, this research moved beyond simply generating topological shapes, explicitly quantifying the fluid dynamic performance of the optimized manifolds under high-fidelity commercial turbulence closures.

Addressing the **first research question**, the theoretical limitations and computational advantages of the Spalart-Allmaras (SA) turbulence model were assessed. While its one-equation, wall-resolved formulation makes it highly tractable for continuous adjoint frameworks, its reliance on the linear Boussinesq approximation fundamentally limits its accuracy in resolving anisotropic Reynolds stresses—particularly in regimes dominated by high streamline curvature and separation.

Regarding the **second research question**, the numerical integration of this highly non-linear SA model into the FEniCS finite-element environment was successfully achieved. By deploying a rigorously decoupled Picard iteration scheme, an Incremental Pressure Correction Scheme (IPCS) for the Navier-Stokes equations, and a stabilized relaxed Eikonal equation for wall-distance computation, the forward fluid state was robustly driven to equilibrium without requiring artificial numerical diffusion.

Building on this foundation, the mathematical framework for topology optimization was constructed. The implementation confirmed that the "frozen turbulence" adjoint assumption—which treats the turbulent eddy viscosity as a static field during gradient assembly—provides a highly stable and computationally efficient optimization engine. Attempts to utilize a fully coupled "semi-frozen" adjoint, which explicitly captures turbulence production sensitivities, revealed significant practical limitations. While theoretically superior, the continuous exact adjoints for the turbulence transport equations proved highly susceptible to numerical noise at the fluid-solid interfaces, frequently causing the optimizer to prioritize localized mathematical artifacts over global fluid routing.

Finally, answering the **third research question**, the framework was subjected to strict cross-code CFD validation. The comparative analysis in Appendix **??** confirmed the fundamental engineering necessity of this thesis: optimizing internal manifolds under turbulent assumptions yields drastically lower power dissipation at high Reynolds numbers compared to baseline laminar designs, primarily by successfully suppressing macroscopic flow separation. Furthermore, the FEniCS-SA topologies demonstrated robust cross-code agreement when re-analyzed as sharp, body-fitted CAD geometries using the Ansys-SA solver.

However, re-evaluating these geometries using the higher-fidelity Ansys SST $k{-}\omega$ model quantified the sensitivity of the optimization to the underlying turbulence physics. In mildly expanding regimes, such as the diffuser and double pipe, the SA-derived topologies performed robustly. Conversely, in the highly curved pipe bend benchmark operating at high Dean numbers, the limitations of the SA optimizer became evident. This confirmed that topologies optimized under isotropic, one-equation assumptions may underperform when subjected to the complex secondary flows and anisotropic shear layers captured by two-equation models.

6.2 Limitations

An objective evaluation of this methodology reveals several important limitations that must be acknowledged:

1. **The Frozen Adjoint Approximation:** By ignoring the derivative of the eddy viscosity with respect to the design field, the optimizer cannot mathematically anticipate future turbulence production. In flows dominated by massive separation, this can blind the optimizer to the downstream hydrodynamic consequences of upstream geometric changes, potentially leading to sub-optimal local minima.

2. **Geometry Extraction Uncertainties:** The transition from a continuous, Brinkman-penalized density field ($\gamma \in [0, 1]$) to a Boolean CAD geometry introduces inherent ambiguity. The specific iso-value threshold chosen to extract the fluid volume dictates the final wall placement. Consequently, the performance discrepancies observed between FEniCS and Ansys are inextricably linked to this manual geometry extraction process, making it difficult to isolate pure solver physics from thresholding effects.

3. **Two-Dimensional Simplification:** The benchmark geometries evaluated in this thesis were restricted to 2D planar domains. Because turbulent phenomena—particularly curvature-induced secondary flows like Dean vortices and turbulent energy cascades—are inherently three-dimensional, the 2D assumption artificially suppresses complex mixing mechanisms that would inevitably occur in physical prototypes.

4. **Lack of Experimental Validation:** While independent cross-code verification against commercial finite-volume solvers strongly corroborates the numerical implementation, it is not a substitute for physical validation. The true operational performance of these algorithmically generated turbulent manifolds requires future validation against experimental pressure-drop and flow-field measurements.

6.3 Future Work

The verified turbulent fluid mechanics framework developed in this thesis establishes a mathematically stable foundation for several advanced avenues of future research:

6.3.1 Conjugate Heat Transfer (CHT) and Thermal Management

The ultimate engineering application for high-Reynolds topology optimization is not merely fluid routing, but active thermal management. Historically, the immense computational cost of resolving Conjugate Heat Transfer (CHT) has forced optimization frameworks to rely on laminar approximations (e.g., natural convection as demonstrated by Alexandersen et al. [3]) or simplified flow proxies such as Darcy models [51]. Even when high-resolution, density-based solvers are successfully deployed for microelectronics cooling, they typically circumvent the complexities of turbulent momentum transport [12].

However, the necessity of capturing true turbulent convection in high-performance heat sinks has driven recent advancements in the literature. Emerging efforts have introduced turbulence into thermal topology optimization using extended finite element (XFEM) level-set approaches with the SA model [33], computationally reduced multi-layer thermofluid approximations [50], and standard density-based frameworks coupled to two-equation $k - \epsilon$ closures [41].

With the turbulent momentum and SA transport equations now stably implemented and verified within the current FEniCS architecture, this framework is perfectly positioned to contribute to this cutting-edge research front. Extending the current methodology to CHT will require implementing a turbulent Prandtl number (Pr_t) approach to model the turbulent heat flux vector, alongside deriving the corresponding thermal adjoint sensitivities to minimize peak temperatures or maximize overall heat transfer coefficients. Furthermore, the capacity for optimizing two-fluid heat exchangers—currently dominated by laminar assumptions [25]—could be significantly enhanced by adapting this turbulent solver.

6.3.2 Three-Dimensional Topologies

As noted in the limitations, turbulent secondary flows are inherently 3D phenomena. To fully evaluate the geometric limits of the turbulence models and design physically realizable manifolds, the current 2D FEniCS implementation must be extended to three dimensions. While the governing equations and adjoint derivations remain mathematically identical, the computational cost of resolving the boundary layers ($y^+ \leq 1$) across a 3D topology optimization grid will require significant advancements in parallel processing and highly efficient linear solvers (e.g., Algebraic Multigrid preconditioners).

6.3.3 Advanced Adjoint Stabilization and Model Hierarchies

Further mathematical research is required to stabilize the monolithic "semi-frozen" adjoint. If the high-frequency numerical artifacts at the domain boundaries can be mitigated—perhaps through advanced filtering techniques or modified boundary penalization—explicitly coupling the turbulence production sensitivities could yield highly efficient geometries in massively separated flows. Additionally, while this thesis relied on the SA model for optimization, future frameworks should investigate the feasibility of utilizing higher-fidelity two-equation models (such as $k-\omega$) directly within the continuous adjoint topology optimization loop to mitigate the curvature limitations identified in Chapter 5.

Appendices

Appendix A

Configuration and Grid Independence of the Spalart-Allmaras Verification Cases

To ensure complete scientific reproducibility and to prevent the disruption of the narrative flow in Chapter 3, all comprehensive configuration matrices, domain truncation proofs, and spatial grid independence studies for the fluid benchmarks are documented in this appendix.

A.1 Verification Benchmark 1: Channel Flow

A.1.1 Physical and Geometric Parameters

The first verification benchmark consists of a pressure-driven periodic 2D channel. To align with fundamental turbulence research conventions, the characteristic length is defined as the full channel height (H), and the Reynolds number (Re_H) is computed strictly based on this geometric parameter rather than the hydraulic diameter. The complete physical setup is detailed in Table A.1.

A.1.2 Numerical Simulation Parameters

The channel flow was resolved using an Incremental Pressure Correction Scheme (IPCS) embedded within a Picard fixed-point iteration loop. The exact numerical solver configurations, convergence tolerances, and relaxation factors utilized in FEniCS are summarized in Table A.2.

A.1.3 Spatial Verification and Grid Independence

To guarantee that the numerical results were independent of the spatial discretization, a formal grid independence study was conducted. Three distinct boundary-

Table A.1: Geometric, physical, and boundary-condition parameters for the periodic channel flow benchmark.

Parameter	**Channel flow setting**
Benchmark type	Pressure-driven, streamwise-periodic 2D channel
Domain dimensions	Length $L = 2.0$ m; full channel height $H = 2.0$ m
Characteristic length	Full channel height $H = 2.0$ m
Reference velocity	$U_{\text{ref}} = 18.5$ m/s, utilized to define the target Reynolds number
Kinematic viscosity	$\nu = 1.81818 \times 10^{-3}$ m^2/s
Reynolds number	$Re_H \approx 2.03 \times 10^4$
Body force	$\mathbf{f} = (0, 0)$ m/s^2
Driving condition	Fixed pressure drop from $p = 2$ Pa to $p = 0$ Pa across the periodic streamwise direction
Velocity boundary conditions	Periodic velocity between inlet/outlet planes; no-slip on top and bottom walls
Pressure boundary conditions	$p = 2$ Pa at the upstream pressure plane; $p = 0$ Pa at the downstream pressure plane
SA boundary conditions	Periodic $\tilde{\nu}$ between inlet/outlet planes; $\tilde{\nu} = 0$ at walls
Initial fields	$\mathbf{u}_0 = (18.5, 0)$ m/s, $p_0 = 2$ Pa, $\tilde{\nu}_0 = 5.0 \times 10^{-3}$ m^2/s
Verification mesh	`Fine_FEniCS`; first layer $\Delta y_1 = 1.709 \times 10^{-3}$ m

layer inflated meshes (Coarse, Medium, and Fine) were evaluated. The geometric properties, node counts, and resolution parameters of this mesh hierarchy are explicitly detailed in Table A.3. Crucially, the first-layer height was kept constant across all grids to strictly enforce the $y^+ \approx 1$ requirement, while the bulk and inflation-layer resolutions were systematically refined.

Visual confirmation of this spatial convergence is provided by the wall-normal velocity profiles plotted in Figure A.1. The `Fine_FEniCS` grid was consequently selected for the final verification against the Ansys Fluent baseline, while the coarser grids demonstrate that the wall-normal velocity profile is already asymptotically converged.

Furthermore, to ensure a rigorous cross-code validation, an equivalent grid independence study was performed for the commercial Ansys Fluent baseline. The Ansys domain was discretized using a comparable mapped mesh hierarchy, maintaining a strict $y^+ \approx 1$ resolution at the walls. The properties of these meshes are documented in Table A.4. The wall-normal velocity profiles extracted from the Ansys solver, illustrated in Figure A.2, exhibited identical asymptotic convergence, confirming that the baseline reference data is mathematically robust and completely independent of spatial discretization errors.

TABLE A.2: Numerical simulation parameters for the FEniCS steady Spalart-Allmaras channel benchmark.

Parameter	**Symbol**	**Setting**
Velocity space	$\mathbf{u}$	Vector CG2, periodic in streamwise direction
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1, periodic in streamwise direction
Quadrature degree	–	2
Maximum outer Picard steps	$N_{\mathrm{Picard,max}}$	200
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-6}
Outer pressure tolerance	ϵ_p	1.0×10^{-6}
Outer SA tolerance	$\epsilon_{\tilde{\nu}}$	1.0×10^{-6}
IPCS pseudo-time step	Δt	5.0×10^{-3} s
Maximum IPCS steps per Picard step	–	500
IPCS velocity tolerance	–	1.0×10^{-6}
IPCS pressure tolerance	–	1.0×10^{-6}
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.3
SA relaxation factor	$\alpha_{\tilde{\nu}}$	0.3
SA lower bound	$\tilde{\nu}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8}; absolute tolerance 1.0×10^{-10}; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: direct MUMPS; SA transport: default solver

TABLE A.3: Wall-resolved channel mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the streamwise resolution, wall-normal resolution, and inflation-layer resolution are refined.

Parameter	**Coarse_FEniCS**	**Medium_FEniCS**	**Fine_FEniCS**
Streamwise cells	10	20	40
Wall-normal cells	42	51	68
First-layer height Δy_1 [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
Minimum wall-normal spacing [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
Maximum wall-normal spacing [m]	7.707×10^{-2}	8.454×10^{-2}	7.589×10^{-2}
Inflation layers per wall	10	14	18
Growth ratio	1.45	1.35	1.25
Vertices	473	1,092	2,829
Triangular elements	840	2,040	5,440

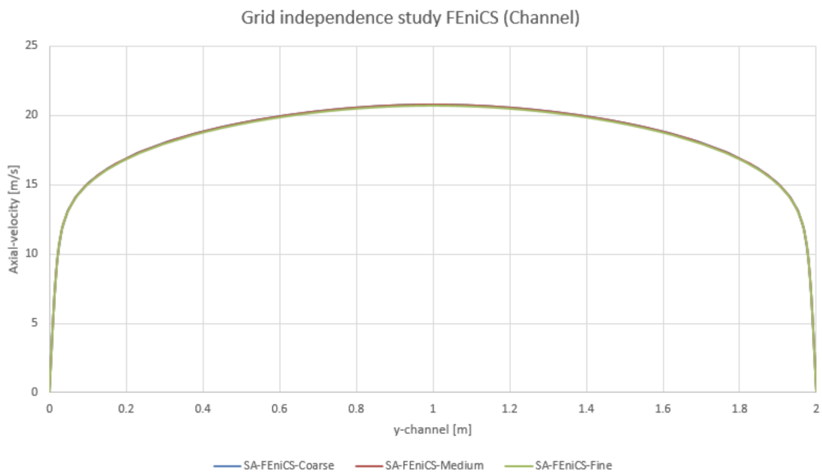


Figure A.1: Wall-normal velocity profiles for the Coarse, Medium, and Fine grids in the periodic channel flow benchmark. The overlapping profiles demonstrate asymptotic spatial convergence.

Table A.4: Wall-resolved channel mesh hierarchy used for the Ansys grid-independence study.

Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Streamwise cells	–	–	–
Wall-normal cells	–	–	–
First-layer height Δy_1 [m]	–	–	–
Minimum wall-normal spacing [m]	–	–	–
Maximum wall-normal spacing [m]	–	–	–
Inflation layers per wall	–	–	–
Growth ratio	–	–	–
Vertices	–	–	–
Elements	–	–	–

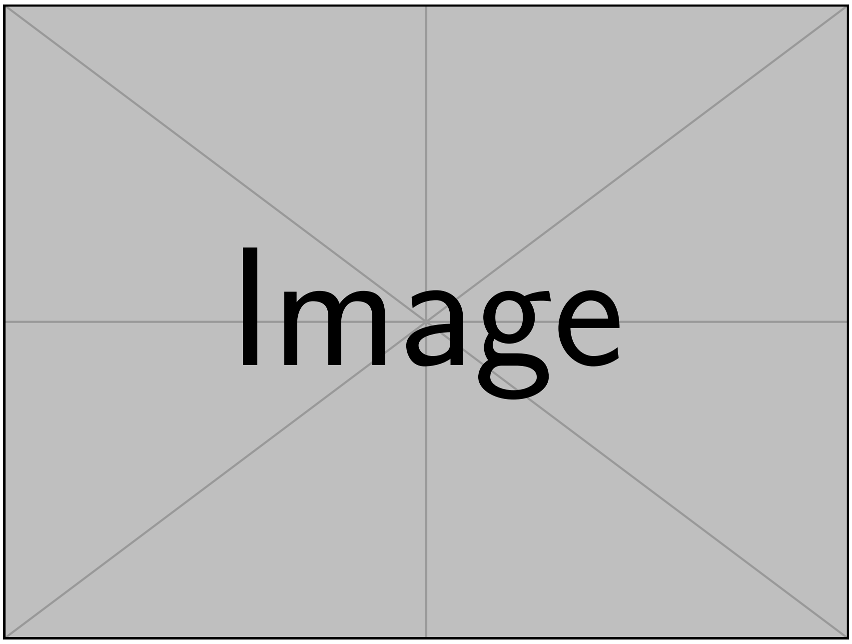


FIGURE A.2: Wall-normal velocity profiles for the Coarse, Medium, and Fine grids in the Ansys periodic channel flow benchmark, demonstrating spatial convergence of the baseline data.

A.2 Verification Benchmark 2: U-Bend Geometry

A.2.1 Physical and Geometric Parameters

The second verification benchmark evaluates a highly convective U-bend. Because this FEniCS domain is a 2D planar approximation of the symmetry mid-plane of the 3D Ansys VMFL048 case, the 3D physical pipe diameter (D) is explicitly retained as the characteristic length scale. This ensures the Reynolds (Re_D) and Dean (De) numbers remain non-dimensionally consistent with the commercial baseline. The configuration is detailed in Table A.5.

Table A.5: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

Parameter	**U-bend setting**
Benchmark type	Truncated 2D symmetry-plane U-bend based on Ansys VMFL048
Domain dimensions	Pipe diameter $D = 0.028$ m; centreline bend radius $R_c = 0.125$ m; straight-leg length $30D = 0.840$ m
Characteristic length	Physical pipe diameter $D = 0.028$ m
Reference / inlet velocity	Uniform prescribed inlet velocity $\mathbf{u}_{in} = (0, -1.42)$ m/s
Kinematic viscosity	$\nu = 8.9 \times 10^{-7}$ m^2/s
Reynolds number	$Re_D \approx 4.47 \times 10^4$
Dean number	$De \approx 1.50 \times 10^4$
Body force	$\mathbf{f} = (0, 0)$ m/s^2
Driving condition	Prescribed inlet velocity and zero relative static pressure at the outlet
Velocity boundary conditions	No-slip on all curved and straight pipe walls; natural velocity condition at pressure outlet
Pressure boundary conditions	$p = 0$ Pa at outlet; pressure left free at inlet and walls
SA boundary conditions	$\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m^2/s at inlet; $\tilde{\nu} = 0$ at walls; natural condition at outlet
Initial fields	$\mathbf{u}_0 = (0, 0)$ m/s, $p_0 = 0$ Pa, $\tilde{\nu}_0 = \tilde{\nu}_{in}$
Verification mesh	`Medium_FEniCS`; first layer $\Delta y_1 = 1.20 \times 10^{-5}$ m

A.2.2 Numerical Simulation Parameters

Due to the adverse pressure gradients and secondary flow separation induced by the streamline curvature, the U-bend represents a significantly more demanding numerical challenge than the channel. Consequently, the FEniCS solver required a smaller pseudo-time step, tighter under-relaxation, and advanced iterative preconditioners to maintain stability, as detailed in Table A.6.

A.2.3 Domain Truncation Study

The original 3D Ansys VMFL048 benchmark utilizes highly extended straight legs measuring 1.555 m to ensure fully developed flow. To reduce computational over-

TABLE A.6: Numerical simulation parameters for the highly convective FEniCS U-bend benchmark.

Parameter	**Symbol**	**Setting**
Velocity space	$\mathbf{u}$	Vector CG2
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1
Quadrature degree	–	4
Maximum outer Picard steps	$N_{\text{Picard,max}}$	180
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-4}
Outer pressure tolerance	ϵ_p	1.0×10^{-4}
Outer SA tolerance	$\epsilon_{\tilde{\nu}}$	1.0×10^{-6}
IPCS pseudo-time step	Δt	1.0×10^{-4} s
Maximum IPCS steps per Picard step	–	180
IPCS velocity tolerance	–	1.0×10^{-4}
IPCS pressure tolerance	–	2.0×10^{-3}
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.1
SA relaxation factor	$\alpha_{\tilde{\nu}}$	0.01
SA lower bound	$\tilde{\nu}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8}; absolute tolerance 1.0×10^{-10}; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: GMRES with ILU velocity/correction preconditioning and Hypre-AMG pressure preconditioning; SA transport: default solver

head for subsequent optimization stages, these legs were truncated to $30D$ (0.840 m). A preliminary sensitivity study was conducted in Ansys Fluent, comparing the full-length and truncated geometries. To mathematically prove that this reduction does not artificially alter the flow physics, the cross-sectional velocity profiles at the critical bend exit were extracted for both domains, as plotted in Figure A.3. The complete overlap of the profiles confirms that the $30D$ truncation successfully isolated the curvature physics without artificially impacting the boundary layer separation.

A.2.4 Spatial Verification and Grid Independence

Managing the highly stretched, non-orthogonal inflation layers around the curved arcs of the U-bend required custom mesh generation via Gmsh. A spatial grid independence study was conducted across three refined unstructured grids. The node counts, element densities, and inflation layer parameters for these generated meshes are explicitly documented in Table A.7. Similar to the channel benchmark, the wall-

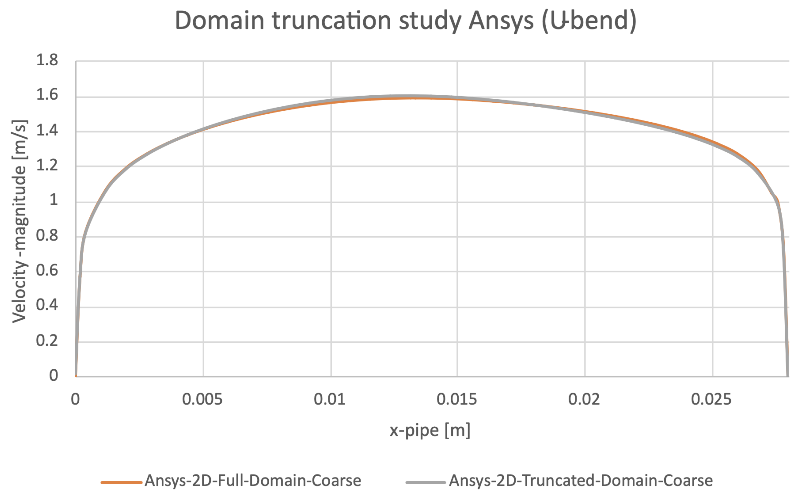


Figure A.3: Velocity profile comparison at the U-bend exit between the full Ansys domain (1.555 m legs) and the truncated computational domain ($30D$ legs). The complete overlap confirms the validity of the truncated geometry.

adjacent element height was strictly maintained at 1.20×10^{-5} m to satisfy the low-Reynolds SA wall constraints, while the internal element sizing was systematically refined.

Table A.7: Wall-resolved U-bend mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the bulk element size and inflation-layer resolution are refined.

Parameter	Coarse_FEniCS	Medium_FEniCS	Fine_FEniCS
Bulk element size [m]	2.80×10^{-3}	1.40×10^{-3}	8.00×10^{-4}
First-layer height Δy_1 [m]	1.20×10^{-5}	1.20×10^{-5}	1.20×10^{-5}
Inflation layers	10	14	18
Growth ratio	1.45	1.30	1.25
Inflation thickness [m]	1.07×10^{-3}	1.53×10^{-3}	2.62×10^{-3}
Vertices	23,751	73,978	182,450
Triangular elements	45,998	144,952	359,646

Convergence was visually verified by extracting the velocity profiles directly at the bend exit, a highly sensitive region for curvature-induced separation, as illustrated in Figure A.4. The chosen `Medium_FEniCS` grid perfectly captured the velocity gradients near the wall and provided sufficient isotropic resolution in the bulk flow to resolve the separation bubbles without numerical instability.

Similarly, to guarantee the integrity of the cross-code validation in this highly convective regime, an equivalent mesh independence study was executed for the Ansys Fluent baseline. Using a parallel unstructured meshing strategy with strict $y^+ \approx 1$ boundary layer inflation (documented in Table A.8), the Ansys velocity

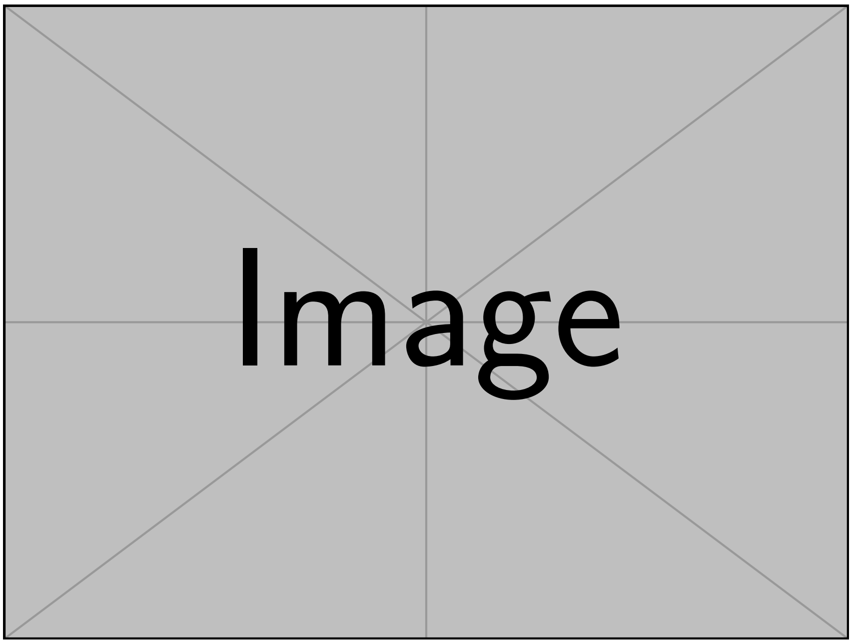


FIGURE A.4: Velocity profiles at the U-bend exit for the Coarse, Medium, and Fine grids in FEniCS. The profiles demonstrate spatial convergence, justifying the selection of the Medium grid for the final verification.

profiles at the bend exit, shown in Figure A.5, were verified to be spatially converged. This ensures that the commercial baseline reference data is independent of grid resolution.

TABLE A.8: Wall-resolved U-bend mesh hierarchy used for the Ansys grid-independence study.

Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Bulk element size [m]	–	–	–
First-layer height Δy_1 [m]	–	–	–
Inflation layers	–	–	–
Growth ratio	–	–	–
Inflation thickness [m]	–	–	–
Vertices	–	–	–
Elements	–	–	–

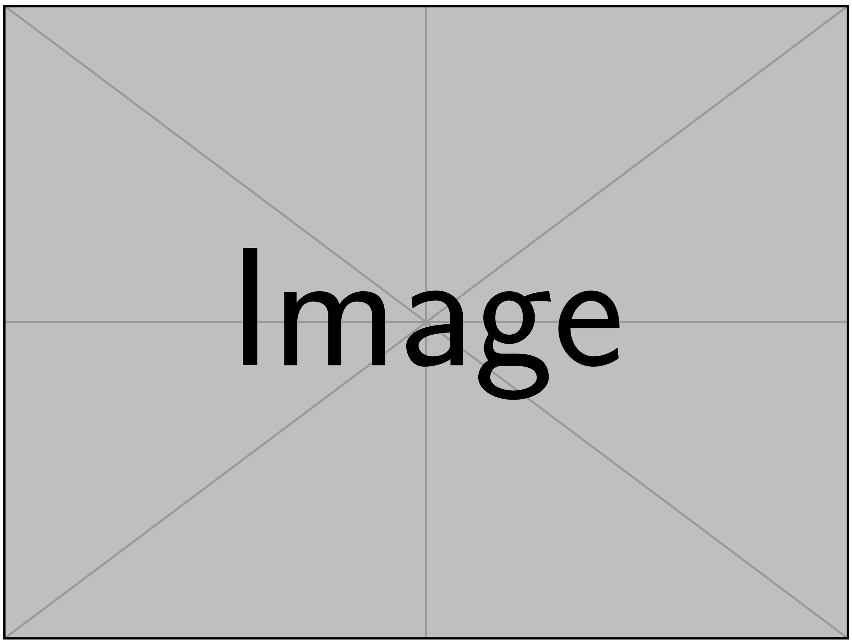


Figure A.5: Velocity profiles at the U-bend exit for the Coarse, Medium, and Fine Ansys grids, demonstrating spatial convergence of the commercial baseline.

Appendix B

Configuration and Grid Independence of the Topology Optimization Benchmarks

To ensure complete scientific reproducibility and support the cross-code validation campaign detailed in Chapter 5, this appendix documents the comprehensive physical boundary conditions, numerical FEniCS configurations, and Ansys spatial grid independence studies for all four topology optimization benchmarks. Because the native FEniCS topology optimization utilizes a fixed, Brinkman-penalized Cartesian grid, the transition to a sharp, body-fitted CAD geometry in Ansys introduces potential spatial discretization errors. Establishing grid independence ensures that the reported viscous dissipation metrics (Φ_D) accurately reflect the underlying solver physics and turbulence models, rather than numerical remeshing noise.

B.1 Near-Wall Treatment and Meshing Strategy

The extracted validation geometries must satisfy the strict near-wall discretization requirements of the selected commercial turbulence closures. Both the Spalart-Allmaras (SA) and SST k-ω models utilized in the Ansys Fluent re-analysis integrate the governing transport equations completely through the viscous sublayer to the solid boundary. This approach strictly circumvents the use of empirical wall functions, necessitating a highly controlled discretization of the boundary layer.

For all evaluated geometries across every benchmark, a hybrid meshing strategy was deployed in Ansys. The bulk flow domains were discretized utilizing unstructured polyhedral elements to efficiently manage complex, optimized internal curves. At all extracted solid boundaries ($\bar{\gamma} = 0$), explicit inflation layers were generated. To guarantee accurate resolution of the severe velocity gradients within the viscous sublayer, the first-layer cell height (Δy_1) was calibrated to maintain a non-dimensional wall distance of $y^+ \leq 1$ across all solid surfaces. Subsequent prism layers were generated with a controlled geometric growth rate not exceeding 1.2 to ensure a smooth volumetric transition into the bulk mesh.

B.2 Benchmark 1: The Diffuser

The diffuser benchmark evaluates the framework's ability to mitigate adverse pressure gradients in expanding flows. The geometric configuration and operational fluid parameters defining this design domain are detailed in Table B.1.

Table B.1: Geometric, physical, and boundary-condition parameters for the Borrvall diffuser benchmark.

Parameter	Diffuser setting
Benchmark type	Borrvall diffuser with one inlet and one pressure outlet
Design domain	Square region $0 \leq x \leq L$, $0 \leq y \leq L$, with $L = 1.0$ m
Non-design fluid regions	Inlet extension $0.2L$ upstream of the design domain; outlet extension $0.2L$ downstream of the design domain
Computational bounds	$-0.2L \leq x \leq 1.2L$, $0 \leq y \leq L$
Outlet opening	Right boundary segment $L/3 \leq y \leq 2L/3$
Characteristic length	$L = 1.0$ m
Density and dynamic viscosity	$\rho = 1.0$ kg/m^3; $\mu = 1.0 \times 10^{-3}$ Pa s
Inlet velocity	Uniform horizontal velocity $\mathbf{u}_{in} = (1.0, 0)$ m/s on the full inlet-extension boundary
Reynolds number	$Re = \rho U_{in} L/\mu = 1.0 \times 10^3$
Outlet condition	Relative static pressure $p = 0$ Pa at the outlet
Wall conditions	No-slip velocity on all walls; $\tilde{\nu} = 0$ at solid walls for the SA model
SA inlet turbulence	Turbulence intensity $I = 5.0\%$; length-scale ratio $\ell/L = 0.07$; inferred $\nu_t/\nu \approx 2.35$
Performance metric	Viscous dissipation over the fluid domain (Φ_D)

To execute the optimization, specific penalty schedules and volume constraints were strictly enforced within the custom finite element solver. These numerical settings are summarized in Table B.2.

Upon convergence, the geometry was extracted and imported into Ansys Fluent. A formal grid independence study was conducted across three distinct spatial resolutions to verify convergence of the target metric. The results are recorded in Table B.3.

As the relative discrepancy between the Medium and Fine meshes ultimately falls below the strict numerical threshold of [Insert %], the **Medium_Ansys** spatial configuration was selected as the reliable standard for the comparative validation.

Table B.2: Numerical FEniCS solver and topology-optimization parameters for the Borrvall diffuser benchmark.

Parameter	**Symbol**	**Setting**
Flow finite elements	$(\mathbf{u}, p)$	Taylor-Hood, Vector CG2 / Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1
Design-density space	γ	Cellwise DG0 with DG Helmholtz density filtering
Reference FEniCS mesh	–	Guided Gmsh mesh, $N = 120$, $h = L/N = 8.33 \times 10^{-3}$ m; 21,492 vertices and 42,406 triangular elements
Adjoint treatment	–	Frozen turbulence; SA field is treated as fixed during the flow adjoint solve
Volume fraction	$V_{\max}$	0.50
RAMP continuation	q	$[0.1, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 3.0]$
Projection continuation	β	$[0.1, 0.5, 1.0, 2.0, 4.0, 8.0, 16.0, 32.0]$ with $\eta = 0.50$
MMA move limits	–	$[0.03, 0.02, 0.015, 0.01, 0.01, 0.0075, 0.005, 0.005]$
Maximum inner iterations	–	$[80, 80, 100, 120, 120, 140, 140, 140]$
Filter radius	r_f	2.0 mesh cells
Quadrature degree	–	6
Forward flow solver	–	SNES Newton line-search solver, `newtonls` with back-tracking
SNES tolerances	–	Relative tolerance 1.0×10^{-6}; absolute tolerance 1.0×10^{-8}; maximum 80 iterations
Frozen NS-SA coupling	–	3 Picard steps; SA relaxation factor 0.05
Linear solver	–	MUMPS
SA wall-distance settings	–	$\sigma_w = 0.01$, $G_0 = 20$, penalty $\alpha_w = 1.0 \times 10^3$, penalty exponent 3, $G_{\min} = 1.0 \times 10^{-8}$

Table B.3: Ansys Fluent mesh-independence table for the extracted Borrvall diffuser geometry.

Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Bulk mesh type	Polyhedral	Polyhedral	Polyhedral
Near-wall treatment	Inflation layers, $y^+ \le 1$	Inflation layers, $y^+ \le 1$	Inflation layers, $y^+ \le 1$
First-layer height Δy_1 [m]	[insert]	[insert]	[insert]
Inflation layers	[insert]	[insert]	[insert]
Growth rate	≤ 1.20	≤ 1.20	≤ 1.20
Elements / cells	[insert]	[insert]	[insert]
Φ_D^{SA} [W/m]	[insert]	[insert]	[insert]
$\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m]	[insert]	[insert]	[insert]
Relative change in Φ_D^{SA} (Medium–Fine)		[insert]%	
Relative change in $\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ (Medium–Fine)		[insert]%	

B.3 Benchmark 2: The Double Pipe

The double pipe benchmark forces complex fluid splitting and mixing. The domain setup, utilizing symmetric inlet and outlet boundary profiles, is formally outlined in Table B.4.

Table B.4: Geometric, physical, and boundary-condition parameters for the Borrvall double-pipe benchmark.

Parameter	**Double-pipe setting**
Benchmark type	Borrvall double pipe with two symmetric inlets and two symmetric pressure outlets
Design domain	Rectangular region $0 \le x \le 1.5L$, $0 \le y \le L$, with $L = 1.0$ m
Non-design fluid regions	$0.2L$ inlet extensions upstream of the two inlet ports; $0.2L$ outlet extensions downstream of the two outlet ports
Computational bounds	$-0.2L \le x \le 1.7L$, $0 \le y \le L$
Port geometry	Port width $L/6$; bottom port $L/6 \le y \le L/3$; top port $2L/3 \le y \le 5L/6$
Characteristic length	Port width $w_p = L/6$
Density and dynamic viscosity	$\rho = 1.0$ kg/m^3; $\mu = 1.0 \times 10^{-4}$ Pa s
Inlet velocities	Two uniform horizontal inlet profiles, $\mathbf{u}_{in} = (1.0, 0)$ m/s at both left ports
Reynolds number	$Re = \rho U_{in} w_p / \mu \approx 1.67 \times 10^3$
Outlet condition	Relative static pressure $p = 0$ Pa at both right ports
Wall conditions	No-slip velocity on all walls; $\tilde{\nu} = 0$ at solid walls for the SA model
SA inlet turbulence	Turbulence intensity $I = 7.5\%$; length-scale ratio $\ell / w_p = 0.06$; inferred $\nu_t / \nu \approx 5.03$
Performance metric	Viscous dissipation over the fluid domain (Φ_D)

The optimization engine relied on the frozen turbulence adjoint subject to a restricted fluid volume fraction. The continuous Galerkin function spaces and penalty tuning parameters are provided in Table B.5.

Following extraction, the resulting CAD manifold was meshed with boundary layer inflation strategies. The macroscopic viscous dissipation convergence across successive mesh refinements is verified in Table B.6.

As the relative discrepancy between the Medium and Fine meshes ultimately falls below the strict numerical threshold of [Insert %], the **Medium_Ansys** spatial configuration was selected as the reliable standard for the comparative validation.

TABLE B.5: Numerical FEniCS solver and topology-optimization parameters for the Borrvall double-pipe benchmark.

Parameter	**Symbol**	**Setting**
Flow finite elements	$(\mathbf{u}, p)$	Taylor-Hood, Vector CG2 / Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1
Design-density space	γ	Cellwise DG0 with DG Helmholtz density filtering
Reference FEniCS mesh	–	Guided Gmsh mesh, $N_x = 150$, $N_y = 100$, $h = 0.01$ m; 27,738 vertices and 54,682 triangular elements
Adjoint treatment	–	Frozen turbulence; SA field is treated as fixed during the flow adjoint solve
Volume fraction	$V_{\max}$	1/3
RAMP continuation	q	$[0.05, 0.10, 0.20, 0.50, 1.00, 1.50, 2.00, 3.00, 3.00, 3.00]$
Projection continuation	β	$[0.5, 1.0, 2.0, 4.0, 8.0, 12.0, 16.0, 24.0, 32.0, 64.0]$ with $\eta = 0.50$
MMA move limits	–	$[0.08, 0.06, 0.04, 0.025, 0.015, 0.008, 0.004, 0.002, 0.001, 0.0005]$
Maximum inner iterations	–	$[50, 80, 80, 80, 100, 120, 140, 160, 180, 200]$
Filter radius	r_f	2.0 mesh cells
Quadrature degree	–	6
Forward flow solver	–	SNES Newton line-search solver with convection continuation and recovery attempts
SNES tolerances	–	Relative tolerance 1.0×10^{-6}; absolute tolerance 1.0×10^{-8}; maximum 120 base iterations
Frozen NS-SA coupling	–	3 Picard steps; SA relaxation factor 0.35
Linear solver	–	MUMPS
SA wall-distance settings	–	$\sigma_w = 0.01$, $G_0 = 20$, penalty $\alpha_w = 1.0 \times 10^3$, penalty exponent 3, $G_{\min} = 1.0 \times 10^{-8}$

TABLE B.6: Ansys Fluent mesh-independence table for the extracted Borrvall double-pipe geometry.

Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Bulk mesh type	Polyhedral	Polyhedral	Polyhedral
Near-wall treatment	Inflation layers, $y^+ \leq 1$	Inflation layers, $y^+ \leq 1$	Inflation layers, $y^+ \leq 1$
First-layer height Δy_1 [m]	[insert]	[insert]	[insert]
Inflation layers	[insert]	[insert]	[insert]
Growth rate	≤ 1.20	≤ 1.20	≤ 1.20
Elements / cells	[insert]	[insert]	[insert]
Φ_D^{SA} [W/m]	[insert]	[insert]	[insert]
$\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m]	[insert]	[insert]	[insert]
Relative change in Φ_D^{SA} (Medium–Fine)		[insert]%	
Relative change in $\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ (Medium–Fine)		[insert]%	

B.4 Benchmark 3: The Pipe Bend

The pipe bend isolates curvature-driven secondary flow phenomena. The specific geometry defining the 90° deflection and the boundary conditions generating the high Dean numbers are presented in Table B.7.

Table B.7: Geometric, physical, and boundary-condition parameters for the Alexandersen pipe-bend benchmark.

Parameter	**Pipe-bend setting**
Benchmark type	Alexandersen pipe bend with a 90° turn from a horizontal inlet to a vertical pressure outlet
Design domain	Square region $0 \leq x \leq L$, $0 \leq y \leq L$, with $L = 1.0$ m
Non-design fluid regions	Inlet lead of length $0.2L$ upstream of the design domain; outlet lead of length $0.2L$ below the design domain
Computational bounds	$-0.2L \leq x \leq L$, $-0.2L \leq y \leq L$
Port geometry	Inlet height $0.2L$ with $0.7L \leq y \leq 0.9L$; outlet width $0.2L$ with $0.7L \leq x \leq 0.9L$
Characteristic length	Inlet height $h_{in} = 0.2L$
Density and dynamic viscosity	$\rho = 1.0$ kg/m^3; $\mu = 4.0 \times 10^{-5}$ Pa s
Inlet velocity	Uniform horizontal velocity $\mathbf{u}_{in} = (1.0, 0)$ m/s
Reynolds number	$Re = \rho U_{in} h_{in}/\mu = 5.0 \times 10^3$
Outlet condition	Relative static pressure $p = 0$ Pa at the outlet; outlet component constraint $u_x = 0$
Wall conditions	No-slip velocity on all walls; $\tilde{\nu} = 0$ at solid walls for the SA model
SA inlet turbulence	Turbulence intensity $I = 7.5\%$; length-scale ratio $\ell/h_{in} = 0.05$; inferred $\nu_t/\nu \approx 12.6$
Performance metric	Viscous dissipation over the fluid domain (Φ_D)

The numerical architecture utilized to solve this highly convective state within the FEniCS optimization loop is documented in Table B.8.

The spatial discretization of the extracted curvature heavily impacts the Ansys solver physics. The asymptotic stabilization of the viscous dissipation across refined element counts is confirmed in Table B.9.

As the relative discrepancy between the Medium and Fine meshes ultimately falls below the strict numerical threshold of [Insert %], the **Medium_Ansys** spatial configuration was selected as the reliable standard for the comparative validation.

TABLE B.8: Numerical FEniCS solver and topology-optimization parameters for the Alexandersen pipe-bend benchmark.

Parameter	Symbol	Setting
Flow finite elements	$(\mathbf{u}, p)$	Taylor-Hood, Vector CG2 / Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1
Design-density space	γ	Cellwise DG0 with DG Helmholtz density filtering
Reference FEniCS mesh	–	Guided Gmsh mesh with $h_{\max} = 0.007$ m; 26,118 vertices and 51,544 triangular elements
Adjoint treatment	–	Frozen turbulence; SA field is treated as fixed during the flow adjoint solve
Volume fraction	$V_{\max}$	0.25
RAMP continuation	q	$[0.05, 0.10, 0.20, 0.40, 0.80, 1.50, 2.50, 3.00, 3.00]$
Projection continuation	β	$[0.10, 0.25, 0.50, 1.00, 2.00, 4.00, 8.00, 16.00, 24.00]$ with $\eta = 0.50$
MMA move limits	–	$[0.08, 0.07, 0.06, 0.045, 0.03, 0.02, 0.01, 0.005, 0.002]$
Maximum inner iterations	–	$[60, 70, 80, 90, 90, 90, 100, 100, 100]$
Filter radius	r_f	3.0 mesh cells
Quadrature degree	–	6
Forward flow solver	–	SNES Newton line-search solver with startup/continuation convection schedules and recovery attempts
SNES tolerances	–	Relative tolerance 1.0×10^{-6}; absolute tolerance 1.0×10^{-8}; maximum 220 base iterations
Frozen NS-SA coupling	–	3 Picard steps; SA relaxation factor 0.20
Linear solver	–	MUMPS
SA wall-distance settings	–	$\sigma_w = 0.01$, $G_0 = 20$, penalty $\alpha_w = 1.0 \times 10^3$, penalty exponent 3, $G_{\min} = 1.0 \times 10^{-8}$; Newton relaxation candidates $[0.35, 0.20, 0.10, 0.05, 0.02, 0.01]$

TABLE B.9: Ansys Fluent mesh-independence table for the extracted Alexandersen pipe-bend geometry.

Parameter	Coarse_Ansys	Medium_Ansys	Fine_Ansys
Bulk mesh type	Polyhedral	Polyhedral	Polyhedral
Near-wall treatment	Inflation layers, $y^+ \leq 1$	Inflation layers, $y^+ \leq 1$	Inflation layers, $y^+ \leq 1$
First-layer height Δy_1 [m]	[insert]	[insert]	[insert]
Inflation layers	[insert]	[insert]	[insert]
Growth rate	≤ 1.20	≤ 1.20	≤ 1.20
Elements / cells	[insert]	[insert]	[insert]
Φ_D^{SA} [W/m]	[insert]	[insert]	[insert]
$\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m]	[insert]	[insert]	[insert]
Relative change in Φ_D^{SA} (Medium–Fine)		[insert]%	
Relative change in $\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ (Medium–Fine)		[insert]%	

B.5 Benchmark 4: The U-Bend

The U-bend represents the most severe aerodynamic challenge, combining strong curvature with forced momentum reversal. The geometric restrictions placed on the solver to manage flow separation are outlined in Table B.10.

Table B.10: Geometric, physical, and boundary-condition parameters for the Alexandersen U-bend benchmark.

Parameter	**U-bend setting**
Benchmark type	Alexandersen U-bend with inlet and outlet on the left boundary and a forced 180° turn
Design domain	Square region $0 \leq x \leq L$, $0 \leq y \leq L$, with $L = 1.0$ m
Non-design fluid regions	Shared left lead region of length $0.2L$ for the inlet and outlet ports
Computational bounds	$-0.2L \leq x \leq L$, $0 \leq y \leq L$
Port geometry	Top inlet port $0.55L \leq y \leq 0.75L$; bottom outlet port $0.25L \leq y \leq 0.45L$; port height $0.2L$
Fixed non-design solid bar	Thickness $0.10L$; rounded-tip radius $0.05L$; total length $0.70L$; bar occupies $0.45L \leq y \leq 0.55L$ and extends from $x = -0.2L$ to tip $x = 0.50L$
Characteristic length	Port height $h_p = 0.2L$
Density and dynamic viscosity	$\rho = 1.0$ kg/m^3; $\mu = 4.0 \times 10^{-5}$ Pa s
Inlet velocity	Uniform horizontal velocity $\mathbf{u}_{in} = (1.0, 0)$ m/s on the top left port
Reynolds number	$Re = \rho U_{in} h_p / \mu = 5.0 \times 10^3$
Outlet condition	Relative static pressure $p = 0$ Pa at the bottom left outlet; outlet component constraint $u_y = 0$
Wall conditions	No-slip velocity on all walls; $\tilde{\nu} = 0$ at solid walls for the SA model
SA inlet turbulence	Turbulence intensity $I = 7.5\%$; length-scale ratio $\ell / h_p = 0.05$; inferred $\nu_t / \nu \approx 12.6$
Performance metric	Viscous dissipation over the fluid domain (Φ_D)

The mathematical stabilization mechanisms and design constraints required to successfully optimize this separating flow in FEniCS are detailed in Table B.11.

Resolving the shear layers on the inner radius of the extracted U-bend requires dense near-wall meshing. The results of the grid independence evaluation ensuring adequate spatial resolution are listed in Table B.12.

As the relative discrepancy between the Medium and Fine meshes ultimately falls below the strict numerical threshold of [Insert %], the **Medium_Ansys** spatial configuration was selected as the reliable standard for the comparative validation.

TABLE B.11: Numerical FEniCS solver and topology-optimization parameters for the Alexandersen U-bend benchmark.

Parameter	**Symbol**	**Setting**
Flow finite elements	$(\mathbf{u}, p)$	Taylor-Hood, Vector CG2 / Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1
Design-density space	γ	Cellwise DG0 with DG Helmholtz density filtering
Reference FEniCS mesh	–	Guided Gmsh mesh with $h_{\max} = 0.007$ m; 25,653 vertices and 50,475 triangular elements
Adjoint treatment	–	Frozen turbulence; SA field is treated as fixed during the flow adjoint solve
Volume fraction	$V_{\max}$	0.27
RAMP continuation	q	$[0.05, 0.10, 0.20, 0.40, 0.80, 1.50, 2.50, 3.00, 3.00]$
Projection continuation	β	$[0.10, 0.25, 0.50, 1.00, 2.00, 4.00, 8.00, 16.00, 24.00]$ with $\eta = 0.50$
MMA move limits	–	$[0.08, 0.07, 0.06, 0.045, 0.03, 0.02, 0.01, 0.005, 0.002]$
Maximum inner iterations	–	$[60, 70, 80, 90, 90, 90, 100, 100, 100]$
Filter radius	r_f	3.0 mesh cells
Quadrature degree	–	6
Forward flow solver	–	IPCS fractional-step solver with adaptive restart controls
IPCS settings	–	$\Delta t = 7.5 \times 10^{-6}$ s; maximum 300 iterations; velocity tolerance 1.0×10^{-4}; pressure tolerance 2.0×10^{-3}
IPCS relaxation and linear solvers	–	Velocity relaxation 0.12; pressure relaxation 0.04; maximum 5 restarts; BiCGSTAB/ILU for velocity and pressure blocks
Frozen NS-SA coupling	–	3 Picard steps; SA relaxation factor 0.25
Linear solver	–	MUMPS for direct solves outside IPCS blocks
SA wall-distance settings	–	$\sigma_w = 0.01$, $G_0 = 20$, penalty $\alpha_w = 1.0 \times 10^3$, penalty exponent 3, $G_{\min} = 1.0 \times 10^{-8}$

Table B.12: Ansys Fluent mesh-independence table for the extracted Alexandersen U-bend geometry.

Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Bulk mesh type	Polyhedral	Polyhedral	Polyhedral
Near-wall treatment	Inflation layers, $y^+ \leq 1$	Inflation layers, $y^+ \leq 1$	Inflation layers, $y^+ \leq 1$
First-layer height Δy_1 [m]	[insert]	[insert]	[insert]
Inflation layers	[insert]	[insert]	[insert]
Growth rate	≤ 1.20	≤ 1.20	≤ 1.20
Elements / cells	[insert]	[insert]	[insert]
Φ_D^{SA} [W/m]	[insert]	[insert]	[insert]
$\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m]	[insert]	[insert]	[insert]
Relative change in Φ_D^{SA} (Medium–Fine)		[insert]%	
Relative change in $\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ (Medium–Fine)		[insert]%	

Appendix C

Performance Comparison of the Laminar and Turbulent Topologies

While Chapter 5 successfully verifies the computational accuracy of the fully coupled FEniCS-SA framework against commercial CFD solvers, an essential engineering justification underpins this methodology: does optimizing a fluid manifold under turbulent conditions yield a physically superior design compared to a standard laminar baseline? Because continuous turbulent adjoint pipelines demand substantially more computational resources, tighter iterative coupling, and rigorous stabilization compared to their laminar equivalents, this mathematical complexity must translate into a measurable reduction in hydrodynamic losses. This appendix quantifies that performance gap.

C.1 Methodology and Evaluation Pipeline

To conduct a rigorous comparative analysis, all four topology optimization benchmarks detailed in Appendix B were optimized twice utilizing the custom FEniCS framework.

The first design iteration was generated using the steady-state, incompressible laminar Navier-Stokes equations at a highly viscous operational regime ($Re = 1$, effectively Stokes flow). This effectively blinded the optimizer to convective momentum transport, flow separation, and turbulent dissipation. The second design iteration utilized the fully coupled Spalart-Allmaras turbulent methodology operating at the target high Reynolds numbers. All geometric dimensions, fluid properties, and physical boundary conditions used for these optimizations match those strictly defined in Appendix B.

Both resulting continuous design fields were subjected to the identical geometry extraction and thresholding procedure. To evaluate their true operational performance, the sharp CAD manifolds were imported into Ansys Fluent, meshed with wall-resolved inflation layers strictly enforcing $y^+ \leq 1$, and subjected to identical high-Reynolds turbulent boundary conditions utilizing the SST k-ω turbulence model. Because the spatial convergence of the turbulent-optimized geometries was already verified in Appendix B, analogous grid independence studies are provided in this appendix exclusively for the extracted laminar baselines.

C.2 Benchmark 1: The Diffuser

The geometric discrepancy between the laminar and turbulent topologies for the diffuser benchmark directly reflects the distinct physical phenomena captured by the respective adjoint sensitivities. Figure C.1 illustrates the thresholded projected density variables ($\bar{\gamma}$) for both solvers.

Placeholder

Projected Density: Laminar ($Re = 1$)

(A) Laminar-optimized topology ($Re = 1$).

Placeholder

Projected Density: Turbulent

(B) Turbulent-optimized topology (SA).

Figure C.1: Comparison of the final thresholded design topologies for the Borrvall diffuser.

To guarantee a fair high-fidelity CFD comparison, the spatial resolution of the extracted laminar design was formally verified in Ansys Fluent. The grid independence study, ensuring the SST k-ω viscous dissipation metric is decoupled from discretization error, is detailed in Table C.1.

Table C.1: Ansys Fluent mesh-independence table for the extracted laminar diffuser geometry.

Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Bulk mesh type	Polyhedral	Polyhedral	Polyhedral
Near-wall treatment	Inflation, $y^+ \leq 1$	Inflation, $y^+ \leq 1$	Inflation, $y^+ \leq 1$
Elements / cells	[insert]	[insert]	[insert]
$\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m]	[insert]	[insert]	[insert]
Relative change in Φ_D (Med–Fine)		[insert]%	

Evaluating both spatially converged geometries under the target turbulent boundary conditions yields the macroscopic thermodynamic performance metrics summarized in Table C.2.

TABLE C.2: Hydrodynamic performance comparison of the laminar and turbulent diffuser topologies, evaluated under identical turbulent boundary conditions (Ansys SST k-ω).

Design Origin	**Viscous Dissipation (Φ_D) [W/m]**	**Relative Performance**
Laminar TO Formulation ($Re = 1$)	[insert]	Baseline
Turbulent TO Formulation (SA)	[insert]	**-[insert]%**

C.3 Benchmark 2: The Double Pipe

The double pipe benchmark forces complex fluid splitting. Figure C.2 compares the unconstrained, purely viscous routing generated by the laminar solver against the streamlined, momentum-conscious routing of the turbulent framework.

Placeholder

Projected Density: Laminar ($Re = 1$)

(A) Laminar-optimized topology ($Re = 1$).

Placeholder

Projected Density: Turbulent

(B) Turbulent-optimized topology (SA).

FIGURE C.2: Comparison of the final thresholded design topologies for the double pipe.

The spatial convergence of the extracted laminar design, evaluated using the SST k-ω closure, is documented in Table C.3. The Medium grid was selected for the final comparison.

TABLE C.3: Ansys Fluent mesh-independence table for the extracted laminar double-pipe geometry.

Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Bulk mesh type	Polyhedral	Polyhedral	Polyhedral
Near-wall treatment	Inflation, $y^+ \leq 1$	Inflation, $y^+ \leq 1$	Inflation, $y^+ \leq 1$
Elements / cells	[insert]	[insert]	[insert]
$\Phi_D^{\text{SST } k\text{-}\omega}$ [W/m]	[insert]	[insert]	[insert]
Relative change in Φ_D (Med–Fine)		[insert]%	

The resulting dissipation values for both optimized geometries are presented in Table C.4, quantifying the performance penalty incurred by neglecting inertial physics during the optimization phase.

Table C.4: Hydrodynamic performance comparison of the laminar and turbulent double-pipe topologies, evaluated under identical turbulent boundary conditions (Ansys SST k-ω).

Design Origin	**Viscous Dissipation (Φ_D) [W/m]**	**Relative Performance**
Laminar TO Formulation ($Re = 1$)	[insert]	Baseline
Turbulent TO Formulation (SA)	[insert]	**-[insert]%**

C.4 Benchmark 3: The Pipe Bend

Curvature-induced separation dominates the 90° pipe bend. As seen in Figure C.3, the laminar solver ($Re = 1$) favors a sharp, unmitigated corner to minimize pure wall friction, whereas the turbulent solver introduces a heavily filleted inner radius to suppress boundary layer detachment.

Placeholder

Projected Density: Laminar ($Re = 1$)

(a) Laminar-optimized topology ($Re = 1$).

Placeholder

Projected Density: Turbulent

(b) Turbulent-optimized topology (SA).

Figure C.3: Comparison of the final thresholded design topologies for the pipe bend.

The grid independence verification for the laminar pipe bend design in Ansys Fluent is provided in Table C.5.

Evaluating these two disparate geometric strategies under the target turbulent regime yields the dissipation metrics shown in Table C.6. The drastic reduction in form drag highlights the necessity of the fully coupled SA framework for highly convective turning flows.

TABLE C.5: Ansys Fluent mesh-independence table for the extracted laminar pipe-bend geometry.

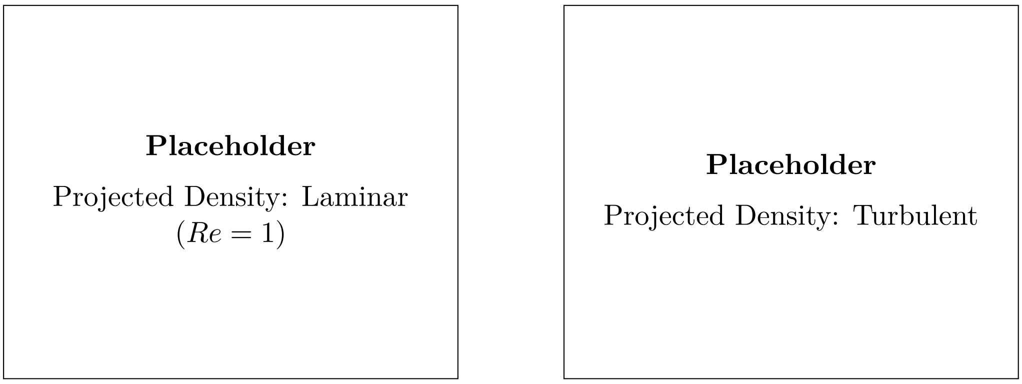
Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Bulk mesh type	Polyhedral	Polyhedral	Polyhedral
Near-wall treatment	Inflation, $y^+ \leq 1$	Inflation, $y^+ \leq 1$	Inflation, $y^+ \leq 1$
Elements / cells	[insert]	[insert]	[insert]
$\Phi_D^{\text{SST } k\text{-}\omega}$ [W/m]	[insert]	[insert]	[insert]
Relative change in Φ_D (Med–Fine)		[insert]%	

TABLE C.6: Hydrodynamic performance comparison of the laminar and turbulent pipe-bend topologies, evaluated under identical turbulent boundary conditions (Ansys SST k-ω).

Design Origin	**Viscous Dissipation (Φ_D) [W/m]**	**Relative Performance**
Laminar TO Formulation ($Re = 1$)	[insert]	Baseline
Turbulent TO Formulation (SA)	[insert]	**-[insert]%**

C.5 Benchmark 4: The U-Bend

The U-bend represents the most severe aerodynamic challenge evaluated in this thesis. Figure C.4 clearly demonstrates the structural differences between a topology optimized purely for laminar diffusion and one actively mitigating massive momentum reversal and turbulent wake generation.



(A) Laminar-optimized topology ($Re = 1$). (B) Turbulent-optimized topology (SA).

FIGURE C.4: Comparison of the final thresholded design topologies for the U-bend.

The spatial convergence of the extracted laminar U-bend CAD was verified to ensure an unbiased comparison. The grid independence parameters are documented in Table C.7.

Table C.8 summarizes the final power dissipation metrics. The severe performance penalty incurred by the laminar geometry is heavily dominated by the artifi-

Table C.7: Ansys Fluent mesh-independence table for the extracted laminar U-bend geometry.

Parameter	**Coarse_Ansys**	**Medium_Ansys**	**Fine_Ansys**
Bulk mesh type	Polyhedral	Polyhedral	Polyhedral
Near-wall treatment	Inflation, $y^+ \leq 1$	Inflation, $y^+ \leq 1$	Inflation, $y^+ \leq 1$
Elements / cells	[insert]	[insert]	[insert]
$\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m]	[insert]	[insert]	[insert]
Relative change in Φ_D (Med–Fine)		[insert]%	

cial turbulent kinetic energy generated within its flow separation zones, confirming that turbulent topology optimization is a strict requirement for high-Reynolds engineering applications.

Table C.8: Hydrodynamic performance comparison of the laminar and turbulent U-bend topologies, evaluated under identical turbulent boundary conditions (Ansys SST k-ω).

Design Origin	**Viscous Dissipation (Φ_D) [W/m]**	**Relative Performance**
Laminar TO Formulation ($Re = 1$)	[insert]	Baseline
Turbulent TO Formulation (SA)	[insert]	**-[insert]%**

Bibliography

[1] A. A. Abuhatira, S. M. Salim, and J. B. Vorstius. Numerical simulation of turbulent pipe flow with 90-degree elbow using wall y+ approach. In *Volume 10: Fluids Engineering*, ASME 2021 International Mechanical Engineering Congress and Exposition. American Society of Mechanical Engineers, 11 2021.

[2] J. Alexandersen. A detailed introduction to density-based topology optimisation of fluid flow problems with implementation in matlab. *Structural and Multidisciplinary Optimization*, 66(1):12, 2023.

[3] J. Alexandersen, N. Aage, C. S. Andreasen, and O. Sigmund. Topology optimisation for natural convection problems. *International Journal for Numerical Methods in Fluids*, 76:699–721, 2014.

[4] J. Alexandersen and C. S. Andreasen. A review of topology optimisation for fluid-based problems. *Fluids*, 5(1):29, 2020.

[5] M. S. Alnæs, J. Blechta, J. Hake, A. Johansson, B. Kehlet, A. Logg, C. Richardson, J. Ring, M. E. Rognes, and G. N. Wells. The fenics project version 1.5. *Archive of Numerical Software*, 3(100), 2015.

[6] D. H. Alonso, L. F. Garcia Rodriguez, and E. C. N. Silva. Flexible framework for fluid topology optimization with openfoam and finite element-based high-level discrete adjoint method (fenics/dolfin-adjoint). *Structural and Multidisciplinary Optimization*, 64(6):4409–4440, 2021.

[7] Altair Engineering, Inc. *AcuSolve Turbulence Modeling Training Manual*, 2023. Available online: https://help.altair.com/hwcfdsolvers/acusolve.

[8] E. Andreassen, A. Clausen, M. Schevenels, B. S. Lazarov, and O. Sigmund. Efficient topology optimization in matlab using 88 lines of code. *Structural and Multidisciplinary Optimization*, 43(1):1–16, 2011.

[9] ANSYS, Inc., Canonsburg, PA. *Ansys Fluid Dynamics Verification Manual, Release 2024 R1*, January 2024.

[10] R. K. Apalowo. Numerical study of different models for turbulent flow in 90° pipe bend. *U.Porto Journal of Engineering*, 8(2):121–134, 2022.

[11] D. N. Arnold, F. Brezzi, B. Cockburn, and L. D. Marini. Unified analysis of discontinuous galerkin methods for elliptic problems. *SIAM Journal on Numerical Analysis*, 39(5):1749–1779, 2002.

[12] B. Barroca, M. Blommaert, G. Buckinx, M. Schevenels, and M. Baelmans. High-resolution topology optimization of heat sinks for microelectronics and benchmark against conventional designs. In *17th International Heat Transfer Conference, IHTC-17*, Cape Town, 2023.

[13] A. Bayat, H. Li, and J. Alexandersen. Density-based topology optimization for turbulent fluid flow using the standard $k-\varepsilon$ RANS model with wall-functions imposed through an implicit wall penalty formulation, Jan. 2026.

[14] M. P. Bendsøe and N. Kikuchi. Generating optimal topologies in structural design using a homogenization method. *Computer methods in applied mechanics and engineering*, 71(2):197–224, 1988.

[15] T. Borrvall and J. Petersson. Topology optimization of fluids in stokes flow. *International journal for numerical methods in fluids*, 41(1):77–107, 2003.

[16] A. J. Chorin. Numerical solution of the navier-stokes equations. *Mathematics of computation*, 22(104):745–762, 1968.

[17] Y. Deng, Z. Liu, and Y. Wu. Topology optimization of steady and unsteady incompressible navier-stokes flows driven by body forces. *Structural and Multidisciplinary Optimization*, 47:555–570, 2013.

[18] C. B. Dilgen, S. B. Dilgen, D. R. Fuhrman, O. Sigmund, and B. S. Lazarov. Topology optimization of turbulent flows. *Computer Methods in Applied Mechanics and Engineering*, 331:363–393, 2018.

[19] M. M. Enayet, M. M. Gibson, A. M. K. P. Taylor, and M. Yianneskis. Laser-doppler measurements of laminar and turbulent flow in a pipe bend. *International Journal of Heat and Fluid Flow*, 3(4):213–219, 1982.

[20] A. Fawaz, Y. Hua, S. Le Corre, Y. Fan, and L. Luo. Topology optimization of heat exchangers: A review. *Energy*, 258:124053, 2022.

[21] J. H. Ferziger, M. Perić, and R. L. Street. *Computational Methods for Fluid Dynamics*. Springer, Berlin, Heidelberg, 4th edition, 2012.

[22] A. Gersborg-Hansen, M. P. Bendsøe, and O. Sigmund. Topology optimization of channel flow problems. *Structural and multidisciplinary optimization*, 30(3):181–192, 2005.

[23] C. Geuzaine and J.-F. Remacle. Gmsh: A 3-d finite element mesh generator with built-in pre-and post-processing facilities. *International journal for numerical methods in engineering*, 79(11):1309–1331, 2009.

[24] J. S. Hesthaven and T. Warburton. *Nodal Discontinuous Galerkin Methods: Algorithms, Analysis, and Applications*. Springer Science & Business Media, New York, NY, 2007.

[25] L. C. Høghøj, D. R. Nørhave, J. Alexandersen, O. Sigmund, and C. S. Andreasen. Topology optimization of two fluid heat exchangers. *International Journal of Heat and Mass Transfer*, 163:120543, 2020.

[26] joove123. k-epsilon fenics implementation. https://github.com/joove123/k-epsilon.git, 2026.

[27] H. Kobayashi, K. Yaji, S. Yamasaki, and K. Fujita. Topology design of two-fluid heat exchange. *Structural and Multidisciplinary Optimization*, 63:821–834, 2021.

[28] H. P. Langtangen and A. Logg. *Solving PDEs in Python: The FEniCS Tutorial I*. Springer Nature, 2016.

[29] B. S. Lazarov and O. Sigmund. Filters in topology optimization based on helmholtz-type differential equations. *International Journal for Numerical Methods in Engineering*, 86(6):765–781, 2011.

[30] A. Logg, K.-A. Mardal, and G. N. Wells. *Automated Solution of Differential Equations by the Finite Element Method: The FEniCS Book*, volume 84. Springer Science & Business Media, 2012.

[31] J. Marcibál. Turbulence modeling in computational fluid dynamics. Master's thesis, University of Southern Denmark, 6 2024.

[32] A.-S. Margetis, E. Papoutsis-Kiachagias, and K. Giannakoglou. The continuous adjoint to the incompressible (d) des spalart-allmaras turbulence models. *Computers & Fluids*, 284:106439, 2024.

[33] L. Noël and K. Maute. Xfem level set-based topology optimization for turbulent conjugate heat transfer problems. *Structural and Multidisciplinary Optimization*, 66(1), 2023.

[34] C. Othmer. Adjoint methods for car aerodynamics. *Journal of Mathematics in Industry*, 4(1):1–25, 2014.

[35] E. M. Papoutsis-Kiachagias and K. C. Giannakoglou. Continuous adjoint methods for turbulent flows, applied to shape and topology optimization: industrial applications. *Archives of Computational Methods in Engineering*, 23(2):255–299, 2016.

[36] S. B. Pope. *Turbulent Flows*. Cambridge University Press, 2000.

[37] C. L. Rumsey et al. Turbulence modeling resource. NASA Langley Research Center, https://turbmodels.larc.nasa.gov, 2023. Accessed: 2024-05-15.

[38] M. Schramm, B. Stoevesandt, and J. Peinke. Optimization of airfoils using the adjoint approach and the influence of adjoint turbulent viscosity. *Computation*, 6(1):10, 2018.

[39] P. Spalart and S. Allmaras. A one-equation turbulence model for aerodynamic flows. *30th aerospace sciences meeting and exhibit*, page 439, 1992.

[40] P. R. Spalart and S. R. Allmaras. A one-equation turbulence model for aerodynamic flows. *La Recherche Aérospatiale*, 1:5–21, 1994.

[41] Y. Sun, M. Hao, and Z. Wang. Topology optimization of turbulent flow cooling structures based on the k–ε model. *Entropy*, 25:1061, 2023.

[42] K. Svanberg. The method of moving asymptotes—a new method for structural optimization. *International journal for numerical methods in engineering*, 24(2):359–373, 1987.

[43] T. Takamasa and M. Kondo. Measurement of gas-liquid two-phase flow in a c-shaped pipe. *Nuclear Engineering and Design*, 154(2):147–156, 1995.

[44] R. Temam. Sur l'approximation de la solution des équations de navier-stokes par la méthode des pas fractionnaires (i). *Archive for Rational Mechanics and Analysis*, 32(2):135–153, 1969.

[45] P. Tucker. Differential equation-based wall distance computation for des and rans. *Journal of Computational Physics*, 190(1):229–248, 2003.

[46] H. K. Versteeg and W. Malalasekera. *An introduction to computational fluid dynamics: the finite volume method*. Pearson education, 2007.

[47] F. M. White and J. Majdalani. *Viscous fluid flow*, volume 3. McGraw-Hill New York, 2006.

[48] H. Wu, D. Wang, and X. Huang. Effect of constant eddy viscosity assumption on optimization using a discrete adjoint method. *Chinese Journal of Aeronautics*, 36(11):102–118, 2023.

[49] G. H. Yoon. Topology optimization for turbulent flow with spalart-allmaras model. *Computer Methods in Applied Mechanics and Engineering*, 303:288–311, 2016.

[50] J. Zhao, M. Zhang, Y. Zhu, R. Cheng, and L. Wang. Topology optimization of turbulent forced convective heat sinks using a multi-layer thermofluid model. *Structural and Multidisciplinary Optimization*, 64(6):3835–3859, 2021.

[51] X. Zhao, M. Zhou, O. Sigmund, and C. S. Andreasen. A "poor man's approach" to topology optimization of cooling channels based on a darcy flow model. *International Journal of Heat and Mass Transfer*, 116:1108–1123, 2018.

[52] T. Zhou, T. Liu, Y. Deng, L. Chen, S. Qian, and Z. Liu. Design of microfluidic channel networks with specified output flow rates using the cfd-based optimization method. *Microfluidics and Nanofluidics*, 21(1), 2017.

[53] A. Zymaris, D. Papadimitriou, K. Giannakoglou, and C. Othmer. Continuous adjoint approach to the spalart–allmaras turbulence model for incompressible flows. *Computers & Fluids*, 38(8):1528–1538, 2009.

Code of conduct and transparency statement on the use of GenAI for KU Leuven-students (academic year 2025–2026)

Generative AI (GenAI) assistance tools can be used to generate various types of content, including text, images, code, video, music, or combinations thereof. Common examples of such tools include ChatGPT, Google Gemini, Microsoft Copilot, Midjourney, Claude.ai, Perplexity.ai, and DALL · E, among others.

This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

Important guidelines and remarks

- **Sensitive or personal data:** Some GenAI tools protect your input and sensitive or personal data better than others. There is often no transparency on what the owners of the AI applications do with the data entered. Therefore, **do not enter sensitive or personal data in free GenAI tools.** More info about what sensitive and personal data are, is described in the three confidentiality levels of the KU Leuven data classification model (non-confidential, confidential, strictly confidential). For confidential data you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter. Try to avoid entering these data. Do so only if strictly necessary, and only to the extent required. For personal data, work with anonymous or pseudonymized data.** Additionally, in case of strictly confidential data this should first be discussed with the teaching staff of the course or your thesis supervisor.

- **Copyrighted materials:** For **lawfully obtained copyrighted material** you should preferably use M**365 Copilot Chat with your KU Leuven**

account. If you use another GenAI tool, you must be **absolutely certain** it does n**ot store or reuse the data you enter.**

- GenAI assistance may not be used for data or topics covered by a Non-Disclosure Agreement (NDA). Even in the absence of an NDA, certain information may still need to be treated as confidential—for example, due to regulatory requirements or the risk of significant harm to the university if disclosed. In case of doubt, check with your teaching staff or supervisor.

- If your master's thesis is under embargo, you should first discuss with your supervisor whether the use of GenAI is permitted.

- Before using a GenAI tool, always consider whether its use is responsible, including from a sustainability perspective (e.g. when using GenAI as a search engine, language assistant,. . .).

- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output, that may not always be correct.

- As a student you are responsible for complying with Article 84 of the Regulations on Education and Examinations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools. **Exam Regulations Article 84:** *"Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity."*

- In order to maintain academic integrity and avoid plagiarism, **more information about being transparent on the use of GenAI assistance and about correctly citing and referencing GenAI** can be found on this website for students (Dutch/English).

- **Additional reading: KU Leuven guidelines on responsible use of Generative AI tools, and other information** (Dutch/English)

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use is

allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: .... **Student number:** ....

Please indicate with "X" whether it relates to a course assignment or to the Bachelor's or Master's thesis:

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Other use (specify here; this may also include a combination of types of use mentioned above):	*add text here*	To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis **beforehand** and explain the intended GenAI use. Also inform the programme director. Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools. Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI.